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STRABLE: Benchmarking Tabular Machine Learning with Strings

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Abstract

Benchmarking tabular learning has revealed the benefit of dedicated architectures, pushing the state of the art. But real-world tables often contain string entries, beyond numbers, and these settings have been understudied due to a lack of a solid benchmarking suite. They lead to new research questions: Are dedicated learners needed, with end-to-end modeling of strings and numbers? Or does it suffice to encode strings as numbers, as with a categorical encoding? And if so, do the resulting tables resemble numerical tabular data, calling for the same learners? To enable these studies, we contribute STRABLE, a benchmarking corpus of 108 tables, all real-world learning problems with strings and numbers across diverse application fields. We run the first large-scale empirical study of tabular learning with strings, evaluating 445 pipelines. These pipelines span end-to-end architectures and modular pipelines, where strings are first encoded, then post-processed, and finally passed to a tabular learner. We find that, because most tables in the wild are categorical-dominant, advanced tabular learners paired with simple string embeddings achieve good predictions at low computational cost. On free-text-dominant tables, large LLM encoders become competitive. Their performance also appears sensitive to post-processing, with differences across LLM families. Finally, we show that STRABLE is a good set of tables to study "string tabular" learning as it leads to generalizable pipeline rankings that are close to the oracle rankings. We thus establish STRABLE as a foundation for research on tabular learning with strings, an important yet understudied area. [1]

1 Introduction: The importance of strings for tabular learning

Benchmarking datasets have been central to progress in machine learning and artificial intelligence: for instance, MNIST [39] and ImageNet [9] drove the deep-learning revolution. Standard benchmarking datasets for a given domain, such as vision or language, enable the emergence of new ideas, sorting out the good from the bad. The model rankings they produce are useful beyond the benchmark itself, as they transfer to new datasets from the same domain [50, 52, 25].

The diversity of data tables, central to enterprise machine learning, may seem like a roadblock to this benchmarking methodology. Yet, tabular data has regularities of its own, such as a strong columnar structure (different quantities and distributions per column [22]). Benchmarks like TALENT [68] or TabArena [12] have guided the development of new tabular-specific deep learning methods —such

[1]Web Page of the Strable benchmark: https://soda-inria.github.io/strable/

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as foundation models [28, 48] and tailored architectures [29, 19] that challenge the dominance of industry standards such as XGBoost [6].

However, datasets in the wild rarely adhere to the strict numerical tables favored by current research; they are frequently populated with string entries, some short — names or codes — while other longer with richer semantic content. Despite this prevalence, the intersection of tabular learning and strings — and the critical challenge of selecting their appropriate representation — remains understudied due to the absence of a solid benchmarking suite. Existing benchmarks often prioritize readily-prepared numerical tables and effectively exclude the high-cardinality and semantic entries found in raw string-heavy tables. To bridge this gap, we need a robust set of datasets capable of supporting empirical work specifically on the domain of tables with strings.

Our contributions are: (i) **STRABLE**, a corpus of 108 real-world tables with raw strings and its string taxonomy; (ii) evidence that modular architectures currently dominate the Pareto frontier and that LLM embeddings need a critical post-processing step to perform well; (iii) an analysis establishing STRABLE's generalizability; (iv) evidence that lightweight encoders suffice on categorical-dominant tables, while large LLMs gain ground for free-text. The paper is organized as follows: Section 2 reviews the benchmarking landscape; Section 3 details STRABLE's construction and taxonomy; Section 4 reports the empirical study; Section 5 analyzes the stability of STRABLE's rankings; Section 6 concludes with the implications of our findings on future research.

2 Context: tabular learning research

2.1 The tabular-learning benchmarking landscape

A need for string tabular learning benchmarks The "iron rule" guiding machine-learning research is to compare pipelines on held-out data [25]. While model rankings remain surprisingly consistent across data splits [52, 50, 25], no algorithm is optimal across all problem classes [67]. Rankings are domain-dependent, and models whose inductive biases match the data distribution perform best [22]. Introducing strings into a table fundamentally changes the data distribution. Can high-cardinality categorical encodings [41, 5] suffice, or do we need models with different inductive biases that leverage string semantics [32]? Can we identify the conditions that make each approach optimal? Answering these questions requires a benchmark that preserves raw, heterogeneous string entries. Existing benchmarks, while rigorous in their respective scopes, were not designed for this purpose. We organize them below by how they handle string features (see also Table B.1).

String-excluding benchmarks Several widely-used benchmark suites focus on numerical or low-cardinality categorical features by design. PMLB [53] explicitly replaces categorical and string-encoded features with numerical integer codes, and PMLBmini [35] inherits this property. OpenML-CC18 [3] filters out high-cardinality features through its post-one-hot feature-count cap. Grinsztajn et al. [22] removes categorical features with more than 20 categories and one-hot-encodes the rest, eliminating high-cardinality string content. TabReD [54] curates datasets that have already undergone feature-engineering, removing raw string content prior to evaluation. TabArena [12] inherits similar curation choices from prior work.

String-flattening benchmarks A larger group of benchmarks retains string columns but converts them to fixed numerical representations before evaluation, preventing the study of alternative string-handling strategies. McElfresh et al. [40], AMLB [18] and TabRepo [55] apply pipeline- or method-specific encodings; AMLB additionally excludes free-form text features at curation time. TALENT [68] and Zabërgja et al. [69] use standard vectorization that treats strings as mathematical vectors, and BeyondArena [47] embeds all text with a single fixed language model. Concurrently, TEmBed [64] benchmarks tabular embeddings across cell, row, column, and table granularities on retrieval and similarity tasks, serializing tables to text for most encoders.

Narrow string-aware benchmarks Few benchmarks evaluate strings, though often with restrictive scopes. Shi et al. [57] pioneer the multimodal tabular setting with 18 text-rich datasets, but many of their tables rely almost entirely on text with only one or two tabular features. CARTE [32] excludes predominantly missing columns and curates datasets with meaningful columns and discrete entries, it has a lower density of text columns that are moderately diverse (Figure E.1). TextTabBench

[43] identifies the mixed-modality gap in tables but focuses only on datasets with semantically rich "free-text" features, and MulTaBench [2] only on tasks where fine-tuned encoders help; both cover a small subset of the tables-with-string domain.

2.2 Progress in tabular learning

Numerical tabular learners For years, Gradient-Boosted Decision Trees such as XGBoost and CatBoost [46] have dominated tabular data, outperforming deep learning on tabular benchmarks [22]. Recent tuned deep baselines like RealMLP [29] and parameter-efficient ensembles like TabM [19] have narrowed this gap, and the emergence of tabular foundation models – TabPFN-2.5 [24] and TabICLv2 [49] – has let in-context learning catch up with tree-based models.

End-to-end string table learners Some end-to-end tabular learners accept tables with strings. Classical methods discard string surface form via static encodings: `CatBoost` treats string columns as categorical features and applies its native ordered target statistics scheme, while `Mambular` [61] ordinally encodes strings and passes them through a learnable embedding lookup with a Mamba backbone. A more recent wave jointly models numbers, string entries, and column names to capture semantics: `CARTE` [32] and `TARTE` [33] pretrain transformers on string embeddings and numerical features; `TabSTAR` [1] unfreezes a language-model encoder for target-aware semantics; and `ConTextTab` [59] combines semantic encoders with cross-row and cross-column attention [36].

3 Building the STRABLE benchmark corpus

Data Collection To construct our benchmark corpus, we aggregated data from 33 distinct sources spanning 8 application fields (Table C.1), ranging from large institutional repositories to community-driven platforms (see Appendix C). The raw data format varied significantly across these domains: while sources like HIFLD (Infrastructure) and ClinicalTrials.gov typically provided structured CSV, other fields like Commerce and Food heavily relied on web-scraped HTML tables or nested JSONs. From the corresponding available datasets, we manually selected tables as follows. First, we only considered tables with at least two string columns and a minimum of 500 samples (limit inspired by [12]). Each table was paired with a target variable for supervised learning. The final corpus comprises 13 binary classification, 19 multi-class classification, and 76 regression tasks.

Minimal Preprocessing We minimize preprocessing to reflect the reality that data preparation is a major bottleneck [8, 60]. Providing raw data enables studying models that automate this burden. Moreover, preprocessing choices (e.g., specific categorical encoders) lack consensus and risk biasing the benchmark against future architectures that might process strings differently. We likewise perform no feature engineering so STRABLE reflects how learners behave on data as-found rather than as-curated. We flattened nested structures, removed duplicate rows, and dropped single-value columns, all-null columns, and rows with missing labels. To prevent leakage, we removed features X_i where X_i is a trivial function of the target. Missing entries were handled by the encoder-learner pipelines. Since the benchmark focuses on I.I.D. evaluation, we adopted a snapshot strategy—extracting only the most recent available year of data. We sub-sampled large tables $(12/108)$ to a maximum of 75,000 rows for all pipelines, leaving rankings unchanged ($\tau = 0.96$, Appendix C.3). For regression tasks, targets often exhibit heavy skew (e.g., wages). While individual applications may benefit from domain-specific loss functions, to ensure fair evaluation across diverse domains we applied a skewness-minimization protocol: from a set of candidate functions—including $\log(y)$, $\log(1+y)$, $\sqrt[3]{y}$, $\operatorname{arcsinh}(y)$, and $\operatorname{sgn}(y) \cdot \log(1+|y|)$ —we selected the transformation that minimized the skewness in the target distribution following [37]. STRABLE's results should therefore be read as a lower bound from which practitioners can measure the added value of their own domain-specific engineering.

Insights into string tables in the wild captured by our corpus STRABLE's column metadata (Figure 1a) follows heavy-tailed distributions: a median of 18 columns, 7.7K rows, 17-character strings per cell, and a median string-column cardinality of 1.2K. While the median number of columns is comparable to OpenML's – a known tabular machine learning repository [62, 4] – STRABLE tables contain roughly 5 times more rows. In addition, STRABLE's strings are shorter and more repetitive than in prior tabular-text studies (Figure E.1) because we include any string type rather

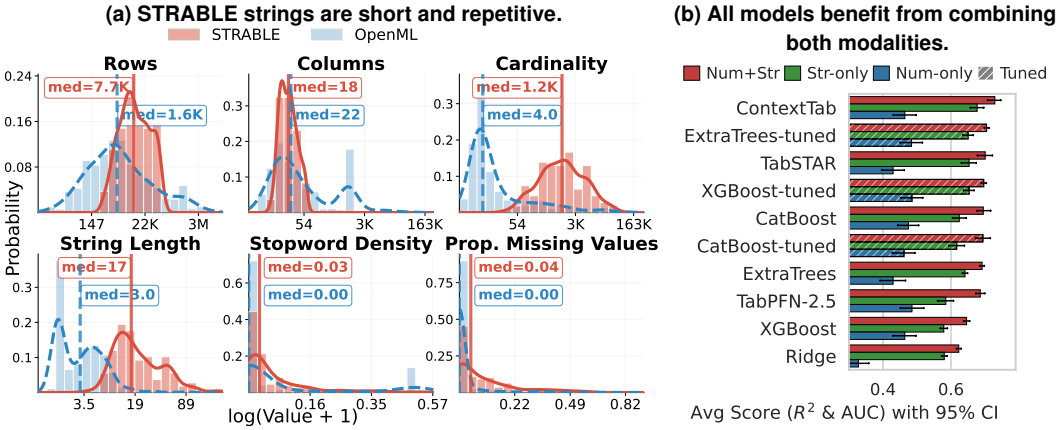


Figure 1: **(a)** STRABLE (solid) vs. OpenML (dashed); cardinality and string length aggregate over string columns. **(b)** Performance for Num-only, Str-only and full table (Num+Str) by learner.

than only long-form text. String columns can be broken into different categories (Appendix C.4): Names (22.78%), Structured Codes (17.01%), Free Text (8.23%), Datetime (1.97%) and Identifiers (0.5%), with the remaining 49.45% being plain Categoricals. Half (50.55%) consist of modalities that string-excluding and string-flattening benchmarks typically ignore or destroy (e.g.: PMLB ignores Names, Structured Codes, Free Text, Datetimes, and Identifiers by collapsing them into integer labels; AMLB removes Free Text columns entirely, Grinsztajn et al. [22] drops categorical features with more than 20 items).

4 Studying today's string tabular learners

Strings carry signal that tabular learning cannot afford to ignore. Indeed, Figure 1b shows that integrating string columns yields a tangible performance improvement across the benchmark for every learner — averaged across encoders for modular pipelines, and across datasets for end-to-end ones — confirming that numeric and string features are complementary. This raises the question of which pipelines best exploit that signal. We run on the STRABLE corpus typical pipelines used today on tables with strings. We benchmark both modular and end-to-end (E2E) architectures.

Modular Pipelines Modular pipelines decompose the learning task into three distinct stages: a string encoder, a dimensionality reduction step for LLMs (post-processing), and a tabular learner.

Encoders. We investigate a high-cardinality categorical encoder, `TargetEncoder` [41], as well as string encoders such as Tf-Idf on character-level n-grams [StringEncoder in 58], or Sentence Transformers [using SBERT, 51]. Appendix D.1 drills down on encoders. We use different LLMs (Table C.3) to embed strings, but focus on the following representative LLMs that were either used by an end-to-end model or appear prominently in the English MTEB benchmark [44]: All-MiniLM-L6-v2 [51], FastText [42], E5-small-v2 [65], LLaMA-3.1-8B [21], Qwen3-Embedding-8B [71], and Jasper-0.6B [70]. Additionally, we include the TARTE encoder [33], which generates row-wise embeddings through pre-trained weights. We generate embeddings for each string cell independently. To obtain a tractable input for tabular learners while preserving relevant signal, these LLM representations are reduced to 30 dimensions.

Post-processing. We evaluate three distinct strategies within this stage: (1) Principal Component Analysis[2] (PCA) on the embeddings of each string column with 30 PC; (2) Standard scaling before PCA, which equalises per-dimension variance, preventing high-variance dimensions from dominating the principal components [30, 56]; (3) Retain the first 30 embedding dimensions (No PCA). While being a natural choice for Matryoshka-trained models [38] like Qwen-3-8B, whose leading dimen-

[2]For Tf-Idf, we rather use an SVD, i.e. we do not center, to avoid densifying the sparse matrix, following Skrub [58].

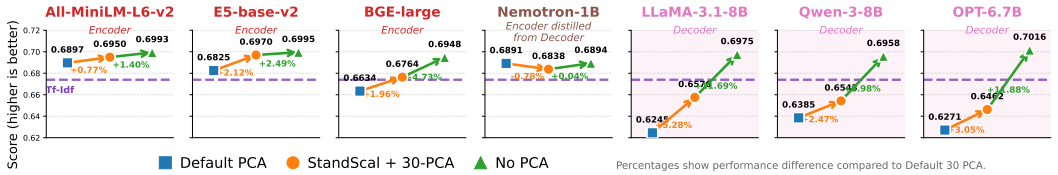


Figure 2: **Post-processing affects LLM-based embeddings, especially for decoder-only models.** Average score across 108 tables and five learners for 7 LM encoders under three post-processing variants. Each panel is one encoder; bars show the mean score under default 30-PCA (blue), standard scaling before 30-PCA (orange), and direct slicing of the first 30 raw embedding dimensions (green). Percentages indicate relative improvement over the default 30-PCA baseline. The dashed purple line marks Tf-Idf. A per-learner breakdown is provided in Figure E.2.

sions are optimized for semantic content, this strategy is applied across all encoders to match the dimensionality of the PCA-based pipelines.

Learners. The resulting numerical tables are used as input to tabular learners of varying sophistication: Ridge [27], Extra-Trees [17], XGBoost, RealMLP, TabM, TabICLv2 and TabPFN-2.5.

E2E architectures We apply on the raw tables end-to-end learners that handle strings without the need of an external encoder: CatBoost[3], Mambular, TabSTAR, and ConTextTab[4].

4.1 Different LLM embeddings hint at different post-processing needs

LLM-based encoders produce high-dimensional embeddings (up to 4096 dimensions for LLaMA-3.1-8B), which require dimensionality reduction before being passed to most tabular learners. The choice of reduction technique affects the downstream score: we compare the three variants and report the average score across five learners: Ridge, XGBoost, ExtraTrees, TabPFN-2.5, TabICLv2 in Figure 2.

Sensitivity to dimensionality reduction. We observe distinct behaviors between model architectures when reducing embeddings to 30 dimensions. **Encoder-only models tolerate PCA reduction.** MiniLM-L6-v2, E5-base-v2 and BGE-large show only marginal gains under standard scaling or raw-dimension slicing, and Nemotron-1B is essentially flat: for these encoders, default PCA is a reasonable choice. Conversely, **decoder-only models hint at greater post-processing sensitivity.** LLaMA-3.1-8B, Qwen-3-8B, and OPT-6.7B all improve when default 30-PCA is replaced either by standard scaling prior to PCA or by retaining the first 30 raw dimensions (six post-processing variants in Figure E.3). Standard scaling rescales each dimension to have unit variance before PCA is applied, making every dimension contribute equally. The performance recovery of `StandScal + PCA` therefore suggests the degradation comes from a few embedding dimensions with large variance dominating the leading principal components. We confirm this empirically: decoder embeddings concentrate most of their variance in a few dimensions, while encoders spread it more evenly (Table E.2, Table E.3). This aligns with a known characteristic of next-token-prediction models: their representations tend to cluster together into a narrow cone [13, 15], which we observe in Figure E.4: LLaMA-3.1-8B and Qwen-3-8B have an average cosine similarity of ≈ 0.57, compared to just 0.25 for MiniLM-L6-v2.

The importance of reduced-dimension embeddings. Our pipelines reduce the embeddings to 30 dimensions, e.g. by PCA. Previous research indicates that larger dimensions offer diminishing returns [23]. Indeed, given the prevalence of short strings in the benchmark (median ≈ 17), high-dimensional representations capture unnecessary complexity. Nevertheless, we vary the pipeline by using 60 and 120 PCA dimensions rather than 30; this yields worse results, increasing the runtime (Figure E.5).

[3]CatBoost falls in this group as we do not couple it with an external encoder; instead, we declare string columns as categorical features and rely on CatBoost's native ordered target statistics.

[4]ConTextTab is pretrained on the T4 web corpus, raising an overlap question with STRABLE; an ablation removing it from the pool (Appendix D.1) shows its inclusion does not affect the benchmark's conclusions.

4.2 Modular pipelines trump today's E2E learners on the Pareto frontier

Modular pipelines set the ceiling for predictive performance. While E2E models are explicitly designed to learn joint representations of heterogeneous data, they are outperformed by modular pipelines on our benchmark. In terms of absolute predictive rank (Figure 3), modular pipelines using post-processed Large Language Models (e.g., Qwen-3-8B) coupled with advanced tabular learners achieve the highest overall performance, consistently outperforming native E2E architectures like TabSTAR and ConTextTab. These E2E models were also shown to be weaker tabular learners than TabPFN on numeric-only tables [12], as also seen in Figure 1b.

Lightweight encoders with advanced learners dominate the Pareto frontier. Pure predictive rankings mask the computational overhead introduced by LLMs. When considering runtime, post-processed LLMs fail to dominate the Pareto frontier (Figure 4). This gap is explained by the composition of the tables in the benchmark: for half of the tables, categorical strings are the leading string type (Figure 6). Consequently, lightweight encoders like Tf-Idf capture the necessary signal at a fraction of the computational cost, pushing them to the sweet spot of the Pareto frontier together with learners like TabICLv2 or TabPFN-2.5.

The simplest models benefit more from heavier encoders. For the simplest learners - linear baselines and ExtraTrees - using sophisticated encoders, such as LLMs, leads to better performance than Tf-Idf. This effect is more visible on average performance (Figure 4a) than on ranks (Figure 3), suggesting gains are not consistent across datasets, but are substantial in magnitude when they occur. Among foundation models, TabPFN-2.5 ranks slightly above TabICLv2 — likely an artifact of pretraining rather than the methods themselves (Table E.11).

5 STRABLE generalizes beyond its specific datasets

5.1 Sufficient datasets to approach oracle rankings

A benchmark should provide model comparisons that generalize to datasets outside of the benchmark. In the following, we quantify the stability of model rankings depending on the number of datasets. To this end, we model our benchmark datasets as sampled from an unknown *population* distribution of datasets. We then want to know how close the model ranking R_N on our benchmark with N datasets is to the oracle ranking R_∞ on the population distribution, or to the model ranking R'_N on a second benchmark with N datasets sampled independently from the population distribution. To model the agreement between rankings, we use the Kendall-τ correlation [31]. For two model rankings, Kendall-τ measures the fraction of pairs of models whose order disagrees in the two rankings:

$$\tau_{N,N} := \tau(R_N, R'_N) = 1 - 2\frac{\#\text{disagreeing model pairs}}{\#\text{model pairs}} \in [-1, 1]\,,$$

and similarly $\tau_{N,\infty} := \tau(R_N, R_\infty)$. Theory (detailed in Appendix A) can relate the agreement to the oracle ranking to agreement between two finite-size benchmarks, $\tau_{N,N}$: for two independent benchmarks R_N, R'_N, their deviations from R_∞ are independent. Hence, asymptotically, we expect twice the disagreement: $\mathbb{E}[1 - \tau_{N,N}] \approx 2\mathbb{E}[1 - \tau_{N,\infty}]$ (exact finite-N factor ≈ 1.41, Appendix A.4).

In other words, a comparison of two finite-size benchmarks accumulates variance from both sides, thus represents a "worst-case" scenario. For any given number of datasets, a benchmark is closer to the truth (the oracle) than it is to another finite benchmark, and the agreement between two independent benchmarks gives a lower bound for the convergence of a single benchmark to the oracle ranking.

In practice, we estimate $\mathbb{E}[\tau_{N,N}]$ by picking disjoint subsets D, D' of STRABLE and averaging over different choices of subsets. As these subsets can be at most 54 datasets, half the size of STRABLE, we derive the asymptotic dependency on the number of datasets theoretically, fit its parameters on subsamples of different sizes $N \leq 54$, extrapolate it to $N = 108$, and then use the theoretical correction to estimate the oracle correlation $\mathbb{E}[\tau_{N,\infty}]$ for $N = 108$. Figure 4b shows the empirical estimate and asymptotic fit. The fitted theoretical agreement to oracle (purple line) illustrates how a single benchmark converges faster to the oracle ranking than the convergence between two benchmarks.

The estimates on Figure 4b show that for two benchmarks of size $N = 54$, $\mathbb{E}[\tau(R, R')] \approx 0.86$, corresponding to 7% disagreeing model pairs. Extrapolated to $N = 108$, we expect $\mathbb{E}[\tau(R, R')] \approx$

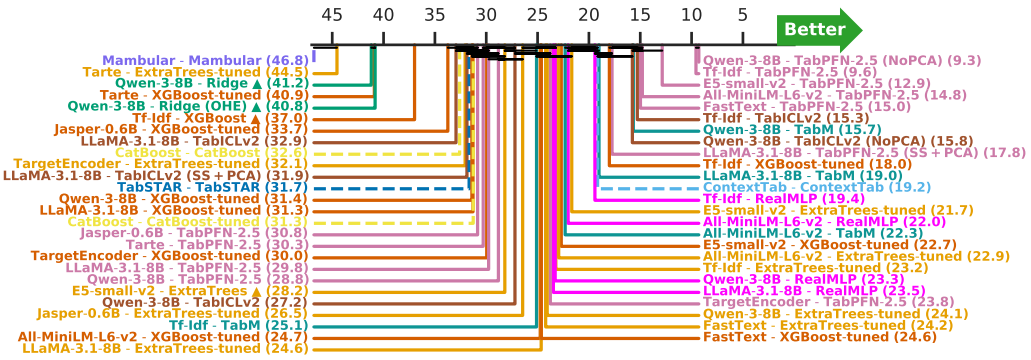


Figure 3: **Critical difference diagram for encoder-learner pipelines.** Pipelines' average rank across the 108 datasets are shown in parentheses; lower is better. Dashed lines are E2E, continuous lines are Modular. Pipelines connected by horizontal bars are not statistically distinguishable at the indicated level (test statistic in Appendix D.1). **Modular pipelines cluster at the top of the ranking.** Pipelines marked with ▲ show only the best-performing encoder for that learner; Figure E.6a details the full set of combinations. Abbreviations: *SS+PCA* = standard scaling before 30-PCA; *NoPCA* = first 30 raw embedding dimensions; *OHE* = one-hot encoding (cardinality threshold 30).

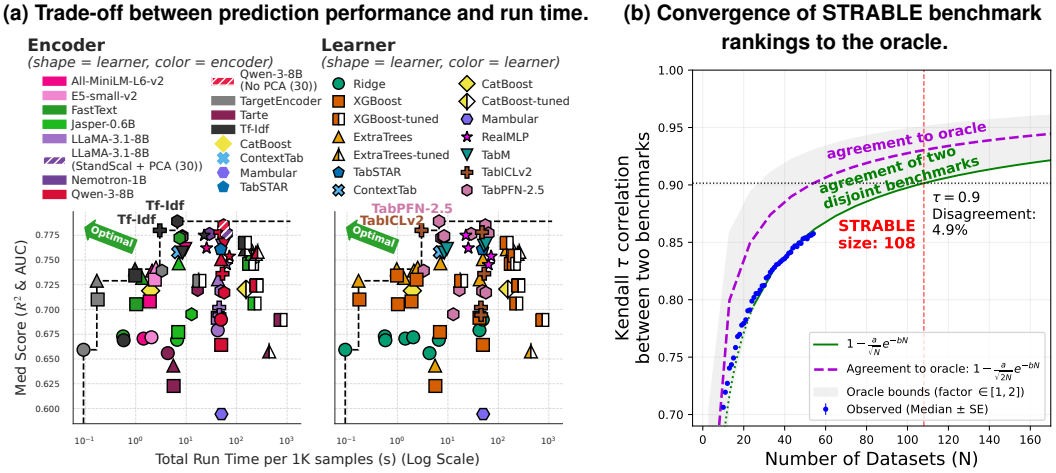


Figure 4: **Pareto-optimality plot and benchmark ranking stability. (a)** Each point is a pipeline, colored by encoder on the left and by learner on the right. The dotted line is the pareto-optimality frontier. **Encoders explain much of the runtime**: for a given encoder, performance varies depending on the learner while runtime varies less (aside from tuning or not). **Simple and advanced learners benefit differently from varying encoders**: for a simple learner such as ridge ●, complex encoders improve prediction performance. More sophisticated learners do not benefit from the most complex encoders (as with TabPFN-2.5 ⬢), unless the encoder is treated with a post-processing method. **(b)** Blue points: observed Kendall-τ between disjoint STRABLE subsets (sub-sampling detailed in Figure E.7). Green curve: fitted asymptotic form. Purple curve: implied convergence to the oracle ranking. Grey band: finite-N correction range (Appendix A.4), reaching $\tau \approx 0.93$ at $N = 108$. **In expectation, a single benchmark is closer to the oracle than to another finite benchmark**, so $\mathbb{E}[\tau]$ between two benchmarks is a **lower bound** on oracle agreement.

0.9, corresponding to 5% disagreeing pairs, whereas for the oracle comparison at $N = 108$ we get $\mathbb{E}[\tau_{N,\infty}] \approx 0.93$, corresponding to 3.5% disagreeing pairs. Overall, this shows that the size of STRABLE allows us to extract model rankings that are close to the oracle.

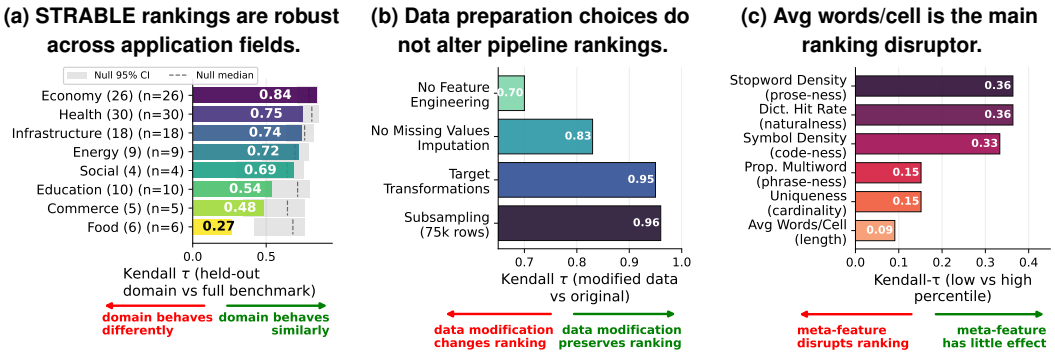


Figure 5: **(a)** Kendall-τ correlation between application-specific subsets and the full benchmark (numbers in parentheses show the number of tables per application field). **(b)** Each row reports Kendall-τ between STRABLE's ranking and the ranking of the opposite data preparation (e.g., applying feature engineering or missing-value imputation, which STRABLE does not; or removing target transformations and subsampling, which STRABLE does). All values are $\tau \approx 0.7$ or higher. **(c)** Kendall-τ between the rankings on the lower and upper 33rd percentile of each string meta-feature; lower τ indicates stronger disruption.

5.2 Pipeline rankings are stable across domains and preprocessing choices

Across application fields To assess generalizability beyond specific domains, we apply a Leave-One-Domain-Out procedure: for each of the 8 categories defined in Table C.1, we measure the Kendall-τ rank correlation between the model ranking on that category alone and the ranking on the full benchmark (Figure 5a). To separate genuine domain effects from sampling noise, we construct a size-matched null reference per domain by drawing $B = 200$ random subsets of size n_D from the full benchmark and computing τ against the full-benchmark ranking. The 95% confidence interval is shown as a grey band behind each bar; its width reflects the statistical power at each domain size. A bar falling inside its band is indistinguishable from a random sample of the same size, and thus representative of the full benchmark. Only Food ($n = 6$, $\tau = 0.27$) falls below its null band, indicating that its high-lexical-diversity product reviews disrupt the ranking more than a random size-6 subset would (domain-level meta-features in Table E.9); Education is borderline ($\tau = 0.54$). All other domains lie within their bands, supporting the conclusion that STRABLE's pipeline ranking is largely domain-independent.

Across data preparation choices Beyond domain selection, benchmark conclusions can also be sensitive to choices in the data-preparation pipeline. We test four such choices, detailed in Section 3: subsampling large tables (default 75K rows vs. full size on 8 of the 12 datasets exceeding the cap), manual feature engineering (raw vs. parsing on 44/108 tables of dates, ranges, coordinates, drug strengths, fiscal periods), target transformation for skewed regression targets (skewness-minimizing transform applied to 61 regression tasks vs. raw target), and missing-value handling (per-learner native handling vs. mean/mode imputation). For each choice, we recompute the full pipeline ranking under an alternative policy and measure its Kendall-τ correlation with the default ranking. Figure 5b summarizes the result: rankings are highly preserved across all four choices ($\tau \in [0.7, 0.96]$), and so are effect sizes: the Tf-Idf-vs-LLM score gap shifts by a median of ≤ 0.005 and at most 0.03 under any policy (Table E.5). Per-policy rankings are reported in Figure E.8–E.11 in the Appendix.

5.3 Dataset features that drive ranking shifts

Section 5.2 established that STRABLE rankings are robust to data preprocessing and applications. We now turn the question around: *what* dataset features cause rankings to shift? We identify these regimes by computing the Kendall-τ between rankings on the upper and lower 33-percentile of six string meta-features.

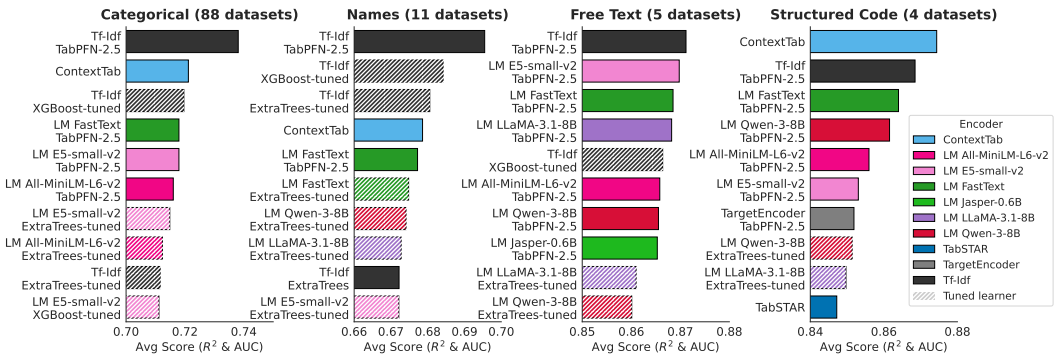


Figure 6: **Top-10 pipelines per leading string type.** Datasets are grouped by their most frequent string type. In the Free Text regime large LLMs enter the top-10 paired with TabPFN-2.5; all other types mirror the global ranking (lightweight encoders at the top paired with TabPFN-2.5, and LM encoders paired with light learners like ExtraTrees). ConTextTab leads the Structured Code panel, plausibly aided by code-rich T4 pretraining tables — though this raises a contamination concern (Appendix D.1). Hatched bars indicate tuned models.

String length is the dominant disruptor of model rankings. We compute six meta-features capturing different facets of "string-ness" (Appendix C.4) and rank them by their ability to disrupt pipeline rankings (Figure 5c). Average words per cell is the single biggest disruptor ($\tau \approx 0.09$), and after it cardinality ($\tau \approx 0.15$): long string cells and cardinality almost entirely overturn the ranking of pipelines. Content-type features (stopword density, dictionary hit rate, symbol density) form a secondary cluster ($\tau \approx 0.33$–0.36).

Free text is the regime where the ranking changes. On tables whose leading string type is Categorical, Names, or Structured Code (Figure 6), the top-10 pipelines mirror the global ranking: Tf-Idf and lightweight LM encoders paired with TabPFN-2.5 dominate. The five tables dominated by Free Text tell a different story: every large LLM (LLaMA-3.1-8B, Qwen-3-8B, Jasper-0.6B) enters the top-10 under default 30-PCA paired with TabPFN-2.5, **indicating that long text carries LLM-accessible signal that TabPFN-2.5 can exploit**. Since most STRABLE string columns are categorical, Tf-Idf suffices for them, while free-text columns benefit from LLM encoders. A cardinality threshold (CT=30) operationalizes this split by routing high-cardinality columns to the LLM and low-cardinality ones to OHE or the learner's native handling (Figure E.12). Columns routed to the LLM under CT=30 contain 2.6× more words per cell (Figure E.13), confirming that the threshold isolates the long-text signal LLM encoders are best equipped to handle.

6 Discussion and Conclusion

STRABLE identifies effective pipelines. We introduce STRABLE to foster research in tabular learning with strings, providing a robust arena of 108 diverse learning problems that embraces the heterogeneity of real-world tables rather than the numerical purity of prior benchmarks. STRABLE provides a robust and stable benchmarking of learning on string tables: the ranking it produces converges toward the oracle ranking ($\tau \approx 0.93$ at $N = 108$) and remains robust to application fields as well as data preparation choices (Section 5.2). Our benchmark yields concrete guidelines for practitioners: lightweight string encoders combined with sophisticated tabular learners (such as TabPFN-2.5) currently dominate the Pareto frontier of performance and runtime, outperforming end-to-end architectures designed for string-tabular learning. LLMs become performance-competitive once their embeddings undergo appropriate post-processing — standard scaling or direct slicing of the first n dimensions — to overcome the bottleneck of traditional PCA.

Refining the role of large LLMs for tabular data with strings. Prior work has reported that larger language models improve performance on tabular data with strings [23]. STRABLE refines this picture along three axes. First, sophisticated encoders deliver consistent gains paired with simple baseline learners (Ridge, ExtraTrees), and — once their embeddings receive appropriate post-processing — become competitive with lightweight encoders even when paired with state-of-

the-art tabular learners (TabPFN-2.5, TabICLv2). Second, STRABLE's strings are short (median 17 characters), which reduces the role of natural language understanding compared to settings with longer free-text fields. Third, decoder-only LLM embeddings are sensitive to default PCA-based dimensionality reduction, which can understate their performance. Together with the heterogeneity findings of Section 5.3, these results suggest that the value of encoding with large LLMs is contingent on the string distribution, the learner they are paired with, and the dimensionality reduction strategy applied to their embeddings.

Limitations. While our work establishes a new standard for this domain, it is not without limitations. STRABLE reflects the string distribution in data-science tables, with short categorical strings rather than long-form text. It thus enables only limited study of sentence-heavy tables. In addition, it does not address time-series specific validation protocols, leaving the exploration of temporal dynamics as well as grouped data for future work.

Looking forward Adapting generative LLMs to tables requires careful alignment of their embedding geometries with tabular learners. Consequently, this research highlights the need for better dimensionality reduction methods for these embeddings. Through this benchmark we hope to catalyze a new wave of research into hybrid architectures that adapt to the diversity of tables, including leveraging semantic understanding when needed.

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Author Contributions

G.B. led the code development, led the experiments, led the writing of the paper, led the evaluation and visualizations, led dataset curation, and led the management of the collaboration. M.J.K. led dataset curation, led the code development and contributed to the experiments. F.L. contributed to the writing of the Appendix A of the paper, to the evaluation, and helped with reviewing and editing. L.P. helped with writing the paper, contributed to the evaluation and visualizations, and helped with reviewing and editing. A.A. helped with writing the paper, contributed to reviewing and editing, and helped with some of the TabSTAR experiments. E.S. contributed to reviewing and editing. R.R. supervised E.S. and A.A. during the project. F.H. supervised L.P. during the project. M.L.M. contributed to writing the paper, contributed to reviewing and editing, helped with the management of the collaboration, and helped with supervising the project, helped with evaluation and visualizations. D.H. contributed to writing the paper, contributed to reviewing and editing, helped with the management of the collaboration, and helped with supervising the project, helped with evaluation and visualizations. G.V. contributed to writing the paper, contributed to reviewing and editing, helped with the management of the collaboration, helped with supervising the project, helped with evaluation and visualizations, secured funding for the project, and supervised G.B. during the project.

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Competing Interests

D.H. is one of the authors of RealMLP. L.P. and F.H. are a subset of the authors of TabPFN-2.6. D.H. and G.V. are a subset of the authors of TabICLv2. L.P. and D.H. are the maintainers of TabArena. G.V. is a core contributor to skrub. G.V. is (co)founder of scikit-learn. L.P., A.A., and F.H. are affiliated with Prior Labs, a company focused on developing tabular foundation models. G.V. is affiliated with probabl, a company focused on developing software for industrial-grade data science. M.J.K. is affiliated to SAP at publication time. The authors declare no other competing interests.

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A Detailed theoretical analysis

A.1 Problem setting and assumptions

We consider the problem of evaluating and ranking k machine learning models, indexed by $i \in \{1, \dots, k\}$, over a population of tabular datasets.

Assumption A.1 (Dataset distribution)**.** There exists an unknown distribution $\mathcal{D}$ of tabular datasets. A benchmark $B = \{d_1, \dots, d_N\}$ consists of N datasets sampled independently and identically distributed (i.i.d.) from $\mathcal{D}$.

Assumption A.2 (Performance decomposition)**.** Let $X_{i,d}$ denote the observed performance metric (e.g., accuracy or R^2 score) of model i on dataset d. We assume this performance can be decomposed into mean performance and deviation:

$$X_{i,d} = \mu_i + \epsilon_{i,d}, \tag{A.1}$$

where $\mu_i \overset{\text{def}}{=} \mathbb{E}_{d\sim\mathcal{D}}[X_{i,d}]$ represents the intrinsic expected performance of model i over the population of datasets.

To facilitate tractable theoretical analysis, we introduce the following assumption regarding the performance deviations.

Assumption A.3 (Homoskedastic Gaussian noise)**.** The deviations $\epsilon_{i,d}$ are independent random variables following a Gaussian distribution with mean zero and constant variance σ^2 across all models and datasets, *i.e.*, $\epsilon_{i,d} \sim \mathcal{N}(0, \sigma^2)$.

Remark A.1 (Limitations). The formulation in Equation A.1 simplifies the problem by treating performance variations as stochastic noise around a global mean μ_i. This basic model has two main limitations: (1) it implies that deviations are purely specific to the instance (i, d); and (2) it ignores structural factors such as dataset difficulty or specific model-dataset affinities (inductive biases). While restrictive, this allows us to derive closed-form analytical bounds on ranking reliability. We relax Assumption A.2 to incorporate explicit inductive biases in subsection A.5.

A.2 Preliminary definitions and results

Let B be a benchmark of size N. We define the observed average performance of model i on benchmark B as:

$$\bar{X}_i^{(B)} \overset{\text{def}}{=} \frac{1}{N} \sum_{d\in B} X_{i,d}. \tag{A.2}$$

We define the observed performance gap between two models i and j on benchmark B as

$$Y_{ij}^{(B)} \overset{\text{def}}{=} \bar{X}_i^{(B)} - \bar{X}_j^{(B)}. \tag{A.3}$$

Using Equation A.1, it can be decomposed as:

$$Y_{ij}^{(B)} = \frac{1}{N} \sum_{d\in B} (X_{i,d} - X_{j,d}) = \Delta_{ij} + \bar{\eta}_{ij}^{(B)}, \tag{A.4}$$

where $\Delta_{ij} \overset{\text{def}}{=} \mu_i - \mu_j$ denotes the expected performance gap, and $\bar{\eta}_{ij}^{(B)} \overset{\text{def}}{=} \frac{1}{N} \sum_{d\in B} (\epsilon_{i,d} - \epsilon_{j,d})$.

Lemma A.1 (Expected sign of the performance gap)**.** *For any pair of models (i, j), the expected sign of the performance gap on a benchmark B of size N is given by*

$$\mathbb{E}\left[\operatorname{sgn}\left(Y_{ij}^{(B)}\right)\right] = \operatorname{sgn}(\Delta_{ij}) \operatorname{erf}\left(\frac{|\Delta_{ij}|\sqrt{N}}{2\sigma}\right), \tag{A.5}$$

where erf *is the error function, defined for all $x \geq 0$ as* $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$.

Proof. Let $p_{ij}(N) \overset{\text{def}}{=} \mathbb{P}\left[\operatorname{sgn}(Y_{ij}^{(B)}) = \operatorname{sgn}(\Delta_{ij})\right]$ denote the probability that the observed ranking of models i and j on benchmark B matches the oracle ranking. The term $\operatorname{sgn}\left(Y_{ij}^{(B)}\right) \operatorname{sgn}(\Delta_{ij})$ takes

the value 1 with probability $p_{ij}(N)$ (when the signs match) and -1 with probability $1 - p_{ij}(N)$ (when they disagree). Therefore, its expected value is:

$$\mathbb{E}\left[\operatorname{sgn}\left(Y_{ij}^{(B)}\right)\operatorname{sgn}(\Delta_{ij})\right] = 1 \cdot p_{ij}(N) + (-1) \cdot (1 - p_{ij}(N)) \tag{A.6}$$

$$= 2p_{ij}(N) - 1. \tag{A.7}$$

Without loss of generality, let us assume that the true gap is positive, i.e., $\Delta_{ij} > 0$. In this case, the probability that the ranking is correct corresponds to the probability that the observed sample mean difference is positive:

$$p_{ij}(N) = \mathbb{P}\left[Y_{ij}^{(B)} > 0\right]. \tag{A.8}$$

Under the homoskedastic independent Gaussian noise assumption (Assumption A.3), we have $\bar{\eta}_{ij}^{(B)} \sim \mathcal{N}\left(0, \frac{2\sigma^2}{N}\right)$, and consequently

$$Y_{ij}^{(B)} \sim \mathcal{N}\left(\Delta_{ij}, \frac{2\sigma^2}{N}\right). \tag{A.9}$$

From Equation A.9, we define the standardized variable Z as:

$$Z \overset{\text{def}}{=} \frac{Y_{ij}^{(B)} - \Delta_{ij}}{\sqrt{\frac{2\sigma^2}{N}}} \sim \mathcal{N}(0, 1). \tag{A.10}$$

The probability of correct ranking becomes:

$$p_{ij}(N) = \mathbb{P}\left[Y_{ij}^{(B)} > 0\right] \tag{A.11}$$

$$= \mathbb{P}\left[Z > -\frac{\Delta_{ij}}{\sqrt{\frac{2\sigma^2}{N}}}\right] \tag{A.12}$$

$$= \Phi\left(\frac{\Delta_{ij}\sqrt{N}}{\sigma\sqrt{2}}\right), \tag{A.13}$$

where Φ is the cumulative distribution function of the standard normal distribution. By symmetry, the result holds for $\Delta_{ij} < 0$ using absolute values:

$$p_{ij}(N) = \Phi\left(\frac{|\Delta_{ij}|\sqrt{N}}{\sigma\sqrt{2}}\right). \tag{A.14}$$

The error function satisfies the identity $\operatorname{erf}(x) = 2\Phi(x\sqrt{2}) - 1$. Using this relation, we can simplify the expectation term $2p_{ij}(N) - 1$ to get

$$2p_{ij}(N) - 1 = \operatorname{erf}\left(\frac{|\Delta_{ij}|\sqrt{N}}{2\sigma}\right). \tag{A.15}$$

Finally, from Equation A.7, we know that $\mathbb{E}\left[\operatorname{sgn}\left(Y_{ij}^{(B)}\right)\right] = \operatorname{sgn}(\Delta_{ij})(2p_{ij}(N) - 1)$, which completes the proof. □

A.3 Agreement between two independent benchmarks

We investigate how the size of a benchmark affects its consistency by asking the following question: given two independent benchmarks B_1 and B_2, both of size N, how well do their model rankings agree? Specifically, we determine how large N must be to achieve a certain level of agreement. To do so, we derive the expected Kendall-τ correlation $\mathbb{E}[\tau_{N,N}]$ between the rankings produced by two independent benchmarks of size N as a function of N.

Let $Y_{ij}^{(B_1)}$ and $Y_{ij}^{(B_2)}$ denote the observed performance gaps between models i and j on benchmarks B_1 and B_2, respectively.

The Kendall-τ correlation between the rankings on B_1 and B_2 is defined as:

$$\tau_{N,N} \stackrel{\text{def}}{=} \frac{2}{k(k-1)} \sum_{i<j} \operatorname{sgn}\left(Y_{ij}^{(B_1)}\right) \operatorname{sgn}\left(Y_{ij}^{(B_2)}\right). \tag{A.16}$$

Proposition A.1 (Expected ranking consistency)**.** *Let $\tau_{N,N}$ denote the Kendall-τ rank correlation between the rankings of k models on two independent benchmarks of size N. The expected correlation is given by*

$$\mathbb{E}[\tau_{N,N}] = \frac{2}{k(k-1)} \sum_{1 \le i < j \le k} \operatorname{erf}^2\left(\frac{|\Delta_{ij}|\sqrt{N}}{2\sigma}\right). \tag{A.17}$$

Proof. Since the benchmarks are independent samples of size N drawn from $\mathcal{D}$, the gaps $Y_{ij}^{(B_1)}$ and $Y_{ij}^{(B_2)}$ are independent and identically distributed (Assumption A.1). Therefore, the expected Kendall-τ is

$$\mathbb{E}[\tau_{N,N}] = \frac{2}{k(k-1)} \sum_{i<j} \mathbb{E}\left[\operatorname{sgn}\left(Y_{ij}^{(B_1)}\right)\right] \mathbb{E}\left[\operatorname{sgn}\left(Y_{ij}^{(B_2)}\right)\right], \tag{A.18}$$

and, from Lemma A.1, we have

$$\mathbb{E}\left[\operatorname{sgn}\left(Y_{ij}^{(B_1)}\right)\right] \mathbb{E}\left[\operatorname{sgn}\left(Y_{ij}^{(B_2)}\right)\right] = \left[\operatorname{sgn}(\Delta_{ij}) \operatorname{erf}\left(\frac{|\Delta_{ij}|\sqrt{N}}{2\sigma}\right)\right]^2, \tag{A.19}$$

$$= \operatorname{erf}^2\left(\frac{|\Delta_{ij}|\sqrt{N}}{2\sigma}\right), \tag{A.20}$$

which completes the proof. □

Equation A.17 shows that $\tau_{N,N}$ converges to 1 as N increases, meaning that ultimately the two benchmarks agree on the ranking. The speed of this convergence is controlled by two parameters: the more separable the models are (large values of $|\Delta_{ij}|$), and the smaller the evaluation noise (small σ), the faster the convergence.

Asymptotic analysis We are interested in the behaviour of the expected Kendall-τ when N becomes large. Indeed, this can help quantify the number of datasets needed to reach a certain degree of agreement between two similarly-sized benchmarks.

Corollary A.1 (Asymptotic agreement rate)**.** *Consider the expected disagreement $1 - \mathbb{E}[\tau_{N,N}]$ between two independent benchmarks of size N. As $N \to \infty$, this disagreement decays exponentially according to the smallest pairwise performance gap:*

$$1 - \mathbb{E}[\tau_{N,N}] \sim \frac{C}{\sqrt{N}} \exp\left(-\frac{\Delta_{\min}^2 N}{4\sigma^2}\right), \tag{A.21}$$

where $\Delta_{\min} \stackrel{\text{def}}{=} \min_{i<j} |\Delta_{ij}|$ is the minimum expected performance gap, and $C = \frac{8\sigma M_{\min}}{k(k-1)\Delta_{\min}\sqrt{\pi}}$ is a constant determined by the number of pairs $M_{\min}$ achieving this minimum gap.

Proof. For large real x, the error function has the following asymptotic expansion

$$\operatorname{erf}(x) = 1 - \frac{e^{-x^2}}{x\sqrt{\pi}} \left(1 + \sum_{n=1}^{\infty} (-1)^n \frac{(2n-1)!!}{(2x^2)^n}\right) \tag{A.22}$$

where $(2n-1)!!$ is the double factorial of $(2n-1)$, which is the product of all odd numbers up to $(2n-1)$.

Therefore, for large x,

$$\operatorname{erf}(x) = 1 - \frac{e^{-x^2}}{x\sqrt{\pi}} + O\left(\frac{e^{-x^2}}{x^3}\right), \tag{A.23}$$

and thus,

$$\operatorname{erf}^2(x) = 1 - \frac{2e^{-x^2}}{x\sqrt{\pi}} + O\left(\frac{e^{-x^2}}{x^3}\right). \tag{A.24}$$

Plugging this into Equation A.17, with $x = \frac{|\Delta_{ij}|\sqrt{N}}{2\sigma}$, we get that for large N

$$\mathbb{E}[\tau_{N,N}] = 1 - \frac{2}{k(k-1)} \sum_{i<j} \left[\frac{4\sigma \exp\left(-\frac{\Delta_{ij}^2 N}{4\sigma^2}\right)}{|\Delta_{ij}|\sqrt{\pi N}} + O\left(\frac{\exp\left(-\frac{\Delta_{ij}^2 N}{4\sigma^2}\right)}{N^{3/2}}\right) \right]. \tag{A.25}$$

Let $\Delta_{\min} = \min_{i<j} |\Delta_{ij}|$. The summation is dominated by the terms corresponding to the smallest performance gap, as the exponential decay is slowest for $\Delta_{\min}$. Therefore, we have for large N:

$$1 - \mathbb{E}[\tau_{N,N}] \sim \frac{C}{\sqrt{N}} \exp\left(-\frac{\Delta_{\min}^2 N}{4\sigma^2}\right) \tag{A.26}$$

with $C = \frac{8\sigma M_{\min}}{k(k-1)\Delta_{\min}\sqrt{\pi}}$, and $M_{\min}$ the number of pairs achieving the minimum gap. □

Corollary A.1 shows that asymptotically the disagreement between benchmarks decays exponentially with N, and is only controlled by the hardest-to-distinguish pair of models.

A.4 Convergence to the oracle ranking

We now address a complementary question: what benchmark size N is required for the empirical ranking of models on $B = \{d_1, \ldots, d_N\}$ to reliably converge to the true oracle ranking over the population $\mathcal{D}$? To quantify this, we examine the expected Kendall-τ correlation, denoted $\mathbb{E}[\tau_{N,\infty}]$, between the empirical ranking observed on a sample of size N and the ground-truth ranking determined by the population expectations $\{\mu_i\}_{i=1}^k$.

The Kendall-τ correlation between the observed ranking on B and the oracle ranking is defined as:

$$\tau_{N,\infty} \overset{\text{def}}{=} \frac{2}{k(k-1)} \sum_{i<j} \operatorname{sgn}\left(Y_{ij}^{(B)}\right) \operatorname{sgn}\left(\Delta_{ij}\right). \tag{A.27}$$

Proposition A.2 (Expected convergence rate to oracle)**.** *Let $\tau_{N,\infty}$ be the Kendall-τ rank correlation between the ranking of k models observed on a benchmark of size N and the oracle ranking. The expected correlation is given by:*

$$\mathbb{E}[\tau_{N,\infty}] = \frac{2}{k(k-1)} \sum_{i<j} \operatorname{erf}\left(\frac{|\Delta_{ij}|\sqrt{N}}{2\sigma}\right). \tag{A.28}$$

Proof. By the linearity of expectation, we have:

$$\mathbb{E}[\tau_{N,\infty}] = \frac{2}{k(k-1)} \sum_{i<j} \mathbb{E}\left[\operatorname{sgn}\left(Y_{ij}^{(B)}\right) \operatorname{sgn}(\Delta_{ij})\right]. \tag{A.29}$$

Combining Lemma A.1 with this equation completes the proof. □

Corollary A.2 (Asymptotic convergence rate to the oracle)**.** *Let $\Delta_{\min} = \min_{i<j} |\Delta_{ij}|$ be the minimum expected performance gap. As $N \to \infty$, the expected disagreement $1 - \mathbb{E}[\tau_{N,\infty}]$ between the ranking on a benchmark of size N and the oracle ranking decays exponentially as:*

$$1 - \mathbb{E}[\tau_{N,\infty}] \sim \frac{C}{2\sqrt{N}} \exp\left(-\frac{\Delta_{\min}^2 N}{4\sigma^2}\right), \tag{A.30}$$

where $C = \frac{8\sigma M_{\min}}{k(k-1)\Delta_{\min}\sqrt{\pi}}$ is the constant defined in Corollary A.1, with $M_{\min}$ representing the number of model pairs separated by exactly $\Delta_{\min}$.

Proof. Substituting the asymptotic expansion of the error function from Equation A.23 into Equation A.28, we obtain:

$$\mathbb{E}[\tau_{N,\infty}] = 1 - \frac{2}{k(k-1)} \sum_{i<j} \left[\frac{2\sigma \exp\left(-\frac{|\Delta_{ij}|^2 N}{4\sigma^2}\right)}{|\Delta_{ij}|\sqrt{\pi N}} + O\left(\frac{\exp\left(-\frac{|\Delta_{ij}|^2 N}{4\sigma^2}\right)}{N^{3/2}} \right) \right]. \tag{A.31}$$

As $N \to \infty$, the summation is dominated by the terms corresponding to the minimal gap $\Delta_{\min}$, as the exponential decay is slowest for these terms. Retaining only the leading order terms yields:

$$1 - \mathbb{E}[\tau_{N,\infty}] \sim \frac{1}{2} \left[\frac{8\sigma M_{\min}}{k(k-1)\Delta_{\min}\sqrt{\pi}} \right] \frac{1}{\sqrt{N}} \exp\left(-\frac{\Delta_{\min}^2 N}{4\sigma^2}\right). \tag{A.32}$$

Identifying the bracketed term as the constant C from Corollary A.1, we recover the result. □

Notably, Corollary A.2 implies that the expected disagreement with the oracle is asymptotically half the disagreement between two independent benchmarks of size N.

Finite-N correction. The relation $\mathbb{E}[1 - \tau_{N,N}] \approx 2\,\mathbb{E}[1 - \tau_{N,\infty}]$ is exact as $N \to \infty$; at finite N, we derive the exact factor relating the two quantities. Using the identity $1 - \mathrm{erf}^2(x_{ij}) = (1 - \mathrm{erf}(x_{ij}))(1 + \mathrm{erf}(x_{ij}))$, this factor is the disagreement-weighted average of $1 + \mathrm{erf}(x_{ij})$ over model pairs, where $x_{ij} = \sqrt{N/2}\,|\hat{\Delta}_{ij}|/\hat{\nu}_{ij}$ is estimated from the mean per-dataset score gaps and their standard deviations (Appendix A.5). Since $0 \le \mathrm{erf}(x) < 1$, the factor lies in $[1, 2]$: it approaches 2 when model pairs are well separated and 1 when they are nearly tied. As the disagreement sum is dominated by near-tied pairs, the factor admits the closed form

$$\frac{\int_0^\infty \left(1 - \mathrm{erf}^2(x)\right) dx}{\int_0^\infty \left(1 - \mathrm{erf}(x)\right) dx} = \sqrt{2} \approx 1.41,$$

which matches the value measured on STRABLE. We use this factor for the oracle curve in Figure 4b, where the grey band spans the full range $[1, 2]$ (i.e., $\mathbb{E}[\tau_{108,\infty}] \in [0.90, 0.95]$). Table A.1 reports the resulting oracle agreement and its difference from the asymptotic factor 2, which shrinks as N grows.

Table A.1: Agreement to the oracle ranking $\mathbb{E}[\tau_{N,\infty}]$ with the finite-N factor $\sqrt{2}$, and its difference from the asymptotic factor 2.

N	$\mathbb{E}[\tau_{N,\infty}]$	difference to factor 2
27	0.863	0.057
54	0.903	0.040
108	0.931	0.028

A.5 Extension: accounting for inductive biases

The simplified model described in Assumption A.2 treats deviations from the mean performance solely as random noise. However, in empirical machine learning, and especially in tabular learning, performance variance is often driven by the structural compatibility between an algorithm's inductive bias and the specific characteristics of a dataset [22]. To capture this heterogeneity, we introduce a latent variable model that accounts for model-dataset affinity.

Assumption A.4 (Inductive bias decomposition)**.** Each dataset $d \in \mathcal{D}$ is associated with a latent vector of meta-features z_d. We decompose the observed performance $X_{i,d}$ into a population mean, an interaction term capturing inductive bias, and residual noise:

$$X_{i,d} = \mu_i + \beta_i^\top z_d + \epsilon_{i,d}, \tag{A.33}$$

where β_i represents the sensitivity vector of model i to the dataset meta-features. This assumption replaces Assumption A.2 for the remainder of this analysis.

Under this formulation, the term $\beta_i^\top z_d$ quantifies the specific affinity between model i and dataset d. To facilitate the derivation of closed-form bounds, we specify the distributional properties of these latent features.

Assumption A.5 (Gaussian meta-features)**.** The latent meta-features are distributed according to a multivariate Gaussian distribution centered at zero, $z_d \sim \mathcal{N}(0, \Sigma_z)$, where Σ_z denotes the covariance of dataset characteristics across the domain $\mathcal{D}$. Furthermore, z_d is independent of the observation noise $\epsilon_{i,d}$.

Remark A.2 (Centering assumption). The centering assumption $\mathbb{E}_d[z_d] = 0$ is made without loss of generality, as any non-zero mean in the latent distribution can be absorbed into the intrinsic performance term μ_i. Indeed, if $\mathbb{E}[z_d] = \bar{z} \neq 0$, we can rewrite Equation A.33 as $X_{i,d} = (\mu_i + \beta_i^\top \bar{z}) + \beta_i^\top (z_d - \bar{z}) + \epsilon_{i,d}$. Defining $\tilde{\mu}_i \overset{\text{def}}{=} \mu_i + \beta_i^\top \bar{z}$ and $\tilde{z}_d \overset{\text{def}}{=} z_d - \bar{z}$, we recover that $\mathbb{E}[\tilde{z}_d] = 0$.

Remark A.3 (Relaxing the Gaussian assumption). We emphasize that the formal assumption of Gaussian latent meta-features z_d serve primarily to simplify exposition. Our following derivations rely solely on the distribution of the benchmark average $\bar{z}_B = \frac{1}{N}\sum_{d \in B} z_d$. Under mild regularity conditions (specifically, the existence of a finite second moment), the central limit theorem ensures that $\bar{z}_B$ converges asymptotically to a Gaussian distribution as $N \to \infty$. Consequently, Assumption A.5 can be relaxed without affecting the asymptotic validity of our results.

Lemma A.2 (Distribution of performance gap under inductive bias)**.** *Under Assumptions A.1, A.3, A.4, and A.5, the observed performance gap $Y_{ij}^{(B)}$ between two models i and j on a benchmark B of size N follows a Gaussian distribution:*

$$Y_{ij}^{(B)} \sim \mathcal{N}\left(\Delta_{ij}, \frac{\nu_{ij}^2}{N}\right), \tag{A.34}$$

where $\nu_{ij}^2 \overset{\text{def}}{=} (\beta_i - \beta_j)^\top \Sigma_z (\beta_i - \beta_j) + 2\sigma^2$ represents the effective pairwise variance, which combines the variance from the observation noise with the one induced by the difference of inductive biases between the models.

Proof. Substituting the decomposition from Equation A.33 into the definition of the performance gap (Equation A.3), we obtain:

$$Y_{ij}^{(B)} = \frac{1}{N}\sum_{d \in B}\left((\mu_i - \mu_j) + (\beta_i - \beta_j)^\top z_d + (\epsilon_{i,d} - \epsilon_{j,d})\right) \tag{A.35}$$

$$= \Delta_{ij} + \gamma_{ij}^\top \bar{z}_B + \bar{\eta}_{i,j}^{(B)}, \tag{A.36}$$

where $\gamma_{ij} \overset{\text{def}}{=} \beta_i - \beta_j$ denotes the differential sensitivity vector, $\bar{z}_B \overset{\text{def}}{=} \frac{1}{N}\sum_{d \in B} z_d$ is the average latent meta-feature vector of the benchmark, and $\bar{\eta}_{i,j}^{(B)}$ is the averaged noise term defined in Equation A.4.

Under Assumption A.5, the linear combination of Gaussian random variables remains Gaussian. Specifically, $\bar{z}_B \sim \mathcal{N}(0, \frac{1}{N}\Sigma_z)$. Since the noise terms ϵ are independent of z (Assumption A.5) and independent across datasets, the terms $\gamma_{ij}^\top \bar{z}_B$ and $\bar{\eta}_{i,j}^{(B)}$ are independent Gaussian variables with zero means. The total variance is thus the sum of their variances:

$$\mathrm{Var}\left(Y_{ij}^{(B)}\right) = \mathrm{Var}\left(\gamma_{ij}^\top \bar{z}_B\right) + \mathrm{Var}\left(\bar{\eta}_{i,j}^{(B)}\right) \tag{A.37}$$

$$= \gamma_{ij}^\top \left(\frac{1}{N}\Sigma_z\right)\gamma_{ij} + \frac{2\sigma^2}{N} \tag{A.38}$$

$$= \frac{1}{N}\left(\gamma_{ij}^\top \Sigma_z \gamma_{ij} + 2\sigma^2\right). \tag{A.39}$$

Defining $\nu_{ij}^2 \overset{\text{def}}{=} \gamma_{ij}^\top \Sigma_z \gamma_{ij} + 2\sigma^2$ yields the result. Note that even if z_d is not strictly Gaussian, the central limit theorem ensures that $\bar{z}_B$ converges to this distribution for sufficiently large N, making the result asymptotically valid under weaker assumptions. □

Starting from Lemma A.2, we can follow the same derivations as in subsection A.3 and subsection A.4 to obtain, under Assumptions A.1, A.3, A.4, and A.5, the four results that follow.

Proposition A.3 (Expected ranking consistency)**.** *Let $\tau_{N,N}$ denote the Kendall-τ rank correlation between the rankings of k models on two independent benchmarks of size N. The expected correlation is given by*

$$\mathbb{E}[\tau_{N,N}] = \frac{2}{k(k-1)}\sum_{1 \leq i < j \leq k} \mathrm{erf}^2\left(\sqrt{\frac{N}{2}} \cdot \frac{|\Delta_{ij}|}{\nu_{ij}}\right). \tag{A.40}$$

Proof. Same as Proposition A.1. □

Corollary A.3 (Asymptotic agreement rate)**.** *Let $\rho_{ij} \overset{def}{=} |\Delta_{ij}|/\nu_{ij}$ denote the signal-to-noise ratio for the pair (i, j), and let $\rho_{\min} \overset{def}{=} \min_{i<j} \rho_{ij}$. As $N \to \infty$, the expected disagreement decays as:*

$$1 - \mathbb{E}[\tau_{N,N}] \sim \frac{\tilde{C}}{\sqrt{N}} \exp\left(-\frac{\rho_{\min}^2 N}{2}\right), \tag{A.41}$$

where $\tilde{C} = \frac{4\sqrt{2}M_{\min}}{k(k-1)\rho_{\min}\sqrt{\pi}}$ and $M_{\min}$ is the number of pairs achieving the minimum ratio $\rho_{\min}$.

Proof. We apply the asymptotic expansion $\operatorname{erf}^2(x) = 1 - \frac{2e^{-x^2}}{x\sqrt{\pi}} + O(x^{-3}e^{-x^2})$ for $x \to \infty$ to Equation A.40, with argument $x_{ij} = \rho_{ij}\sqrt{N/2}$. The summation is dominated by the terms with the slowest exponential decay, which corresponds to the smallest coefficient ρ_{ij}. Unlike the homoskedastic setting in Corollary A.1, where the convergence rate was dictated by the performance gap $\Delta_{\min}$, here it is governed by the signal-to-noise ratio $\rho_{\min}$. This implies that a pair of models with a large performance gap Δ_{ij} may still be the bottleneck for ranking convergence if their relative performance has a high variance (high ν_{ij}) due to different inductive biases (high γ_{ij}). □

Proposition A.4 (Expected convergence rate to oracle)**.** *Let $\tau_{N,\infty}$ be the Kendall-τ rank correlation between the ranking of k models observed on a benchmark of size N and the oracle ranking. The expected correlation is given by:*

$$\mathbb{E}[\tau_{N,\infty}] = \frac{2}{k(k-1)} \sum_{i<j} \operatorname{erf}\left(\sqrt{\frac{N}{2}} \cdot \frac{|\Delta_{ij}|}{\nu_{ij}}\right). \tag{A.42}$$

Proof. Same as Proposition A.2. □

Corollary A.4 (Asymptotic convergence rate to the oracle)**.** *As $N \to \infty$, the expected disagreement $1 - \mathbb{E}[\tau_{N,\infty}]$ between the ranking on a benchmark of size N and the oracle ranking decays as:*

$$1 - \mathbb{E}[\tau_{N,\infty}] \sim \frac{\tilde{C}}{2\sqrt{N}} \exp\left(-\frac{\rho_{\min}^2 N}{2}\right), \tag{A.43}$$

where $\tilde{C}$ and $\rho_{\min}$ are the constants defined in Corollary A.3.

Proof. Applying the asymptotic expansion of the error function $\operatorname{erf}(x) \approx 1 - \frac{e^{-x^2}}{x\sqrt{\pi}}$ to Equation A.42 yields a summation dominated by the terms corresponding to the minimum signal-to-noise ratio $\rho_{\min}$. Comparing this to the expansion of $1 - \mathbb{E}[\tau_{N,N}]$ derived in Corollary A.3, which relies on $1 - \operatorname{erf}^2(x) \approx 2(1 - \operatorname{erf}(x))$, we observe that the expected disagreement with the oracle is asymptotically half the expected disagreement between two independent benchmarks. The exponential decay rate remains controlled by $\rho_{\min}$. □

A.6 Disagreement at position 1: identifying the best model

The Kendall-τ analysis characterises the ranking disagreement across *all* model pairs. A practically more direct question is: how often do two independent benchmarks disagree on *which model ranks first*? We derive a bound on this probability and compare its convergence rate to that of Kendall-τ.

Let $i^* \overset{\text{def}}{=} \arg\max_i \mu_i$ be the oracle best model and $\Delta_1 \overset{\text{def}}{=} \mu_{i^*} - \max_{j \neq i^*} \mu_j > 0$ the margin between the best and second-best model.

Definition A.4 (Position-1 disagreement)**.** Let R_N and R'_N be rankings produced by two independent benchmarks B, B' of size N. The *position-1 disagreement event* is

$$\varepsilon_1(N) \overset{\text{def}}{=} \left\{ \arg\max_i \bar{X}_i^{(B)} \neq \arg\max_i \bar{X}_i^{(B')} \right\}.$$

Proposition A.5 (Position-1 disagreement probability)**.** *Assumptions A.1– A.3,*

$$\mathbb{P}\big[\varepsilon_1(N)\big] \;\le\; 2(k-1)\,\Phi\!\left(-\frac{\Delta_1\sqrt{N}}{\sigma\sqrt{2}}\right),$$

where Φ is the standard normal CDF and k is the number of models.

Proof. **Step 1.** The event $\varepsilon_1(N)$ requires at least one benchmark to rank the wrong model first, so

$$\varepsilon_1(N) \;\subseteq\; \big\{\arg\max_i \bar{X}_i^{(B)} \neq i^*\big\} \;\cup\; \big\{\arg\max_i \bar{X}_i^{(B')} \neq i^*\big\}.$$

Since B and B' are i.i.d., both events have equal probability, and by the union bound:

$$\mathbb{P}[\varepsilon_1(N)] \;\le\; 2\,\mathbb{P}\big[\arg\max_B \neq i^*\big].$$

Step 2. The best model fails to rank first on B iff at least one rival beats it:

$$\big\{\arg\max_i \bar{X}_i^{(B)} \neq i^*\big\} \;=\; \bigcup_{j\neq i^*} \big\{\bar{X}_j^{(B)} > \bar{X}_{i^*}^{(B)}\big\}.$$

A second union bound over the $k-1$ rivals gives

$$\mathbb{P}\big[\arg\max_i \bar{X}_i^{(B)} \neq i^*\big] \;\le\; \sum_{j\neq i^*} \mathbb{P}\big[\bar{X}_j^{(B)} > \bar{X}_{i^*}^{(B)}\big].$$

Step 3. Under Assumption A.3, $\bar{X}_{i^*}^{(B)} - \bar{X}_j^{(B)} \sim \mathcal{N}(\Delta_{i^*j},\, 2\sigma^2/N)$, where $\Delta_{i^*j} = \mu_{i^*} - \mu_j$. Standardising:

$$\mathbb{P}\big[\bar{X}_j^{(B)} > \bar{X}_{i^*}^{(B)}\big] \;=\; \Phi\!\left(-\frac{\Delta_{i^*j}\sqrt{N}}{\sigma\sqrt{2}}\right).$$

Since $\Delta_{i^*j} \ge \Delta_1$ for all $j \neq i^*$, and Φ is increasing, a larger gap yields a more negative argument and thus a smaller value:

$$\Phi\!\left(-\frac{\Delta_{i^*j}\sqrt{N}}{\sigma\sqrt{2}}\right) \;\le\; \Phi\!\left(-\frac{\Delta_1\sqrt{N}}{\sigma\sqrt{2}}\right).$$

Summing over $k-1$ rivals and combining with Steps 1–2 gives the result. □

Corollary A.5 (Asymptotic decay of position-1 disagreement)**.** *As $N \to \infty$,*

$$\mathbb{P}[\varepsilon_1(N)] \;\le\; C_1 \cdot N^{-1/2} \cdot \exp\!\left(-\frac{\Delta_1^2 N}{4\sigma^2}\right), \qquad C_1 = \frac{2(k-1)\,\sigma}{\Delta_1\sqrt{\pi}},$$

obtained from the Gaussian tail bound $\Phi(-t) \le e^{-t^2/2}/(t\sqrt{2\pi})$ [63] applied with $t = \Delta_1\sqrt{N}/(\sigma\sqrt{2})$.

Comparison with Kendall-τ convergence. Corollary A.1 showed that Kendall-τ disagreement decays with exponent $-\Delta_{\min}^2/(4\sigma^2)$, where $\Delta_{\min} = \min_{i<j}|\Delta_{ij}|$ is the smallest gap over *all* model pairs. Since Δ_1 is a gap between i^* and its nearest rival, whereas $\Delta_{\min}$ is the smallest gap between any two models in the entire set (which may be achieved by two closely matched mid-ranking models rather than by the top pair) we have $\Delta_1 \ge \Delta_{\min}$. Setting each bound equal to a threshold ε and solving, the required benchmark sizes satisfy

$$N^*_{\text{pos-1}} \;\propto\; \frac{4\sigma^2}{\Delta_1^2}\log\frac{1}{\varepsilon}, \qquad N^*_{\text{Kendall}} \;\propto\; \frac{4\sigma^2}{\Delta_{\min}^2}\log\frac{1}{\varepsilon}, \qquad \frac{N^*_{\text{pos-1}}}{N^*_{\text{Kendall}}} = \left(\frac{\Delta_{\min}}{\Delta_1}\right)^2 \le 1.$$

Position-1 identification is therefore guaranteed at a smaller benchmark size than full ranking stability, by a factor of $(\Delta_{\min}/\Delta_1)^2$.

Remark A.5 (Tightness of the bound). The union bound in Proposition A.5 is not tight: it treats all $k-1$ rivals as if each had probability $\Phi(-\Delta_1\sqrt{N}/(\sigma\sqrt{2}))$ of beating i^*, whereas rivals with larger gaps $\Delta_{i^*j} \gg \Delta_1$ contribute negligibly. The bound suffices to establish the exponential decay rate and the qualitative comparison with Kendall-τ.

A.7 Top-1 disagreement under inductive biases

Section A.6 assumed that the difficulty of distinguishing i^* from a rival j is determined solely by their mean gap Δ_{i^*j}, with all model pairs sharing the same noise level σ. This assumption breaks down when models have inductive biases: systematic tendencies to perform well on certain dataset types and poorly on others. Two models with different inductive biases will have a performance difference that varies *systematically* across datasets, not just randomly.

Lemma A.2 showed that the sample average difference satisfies

$$\bar{X}_{i^*}^{(B)} - \bar{X}_j^{(B)} \;\sim\; \mathcal{N}\left(\Delta_{i^*j},\; \frac{\nu_{i^*j}^2}{N}\right),$$

where $\nu_{i^*j}^2 = (\beta_{i^*} - \beta_j)^\top \Sigma_z (\beta_{i^*} - \beta_j) + 2\sigma^2$. The first term captures inductive bias variability: it is large when i^* and j respond differently to dataset characteristics ($\beta_{i^*} \neq \beta_j$) and when the benchmark spans diverse dataset types (large Σ_z). The second term $2\sigma^2$ is the pure noise contribution from Section A.6. Critically, $\nu_{i^*j}^2$ differs across pairs: a rival whose biases mirror i^*'s recovers $\nu_{i^*j}^2 \approx 2\sigma^2$, while a rival with very different biases has $\nu_{i^*j}^2 \gg 2\sigma^2$. The probability that rival j beats i^* on benchmark B is $\Phi(-\Delta_{i^*j}\sqrt{N}/\nu_{i^*j})$, which depends on Δ_{i^*j} and ν_{i^*j} only through their ratio $\rho_{i^*j} = \Delta_{i^*j}/\nu_{i^*j}$. A large mean gap Δ_{i^*j} does not therefore guarantee a small error probability if ν_{i^*j} is comparably large: what governs the difficulty of beating rival j is whether the gap is large *relative to* the variability of their performance difference across datasets.

Definition A.6 (Per-pair SNR for top-1)**.** For each rival $j \neq i^*$, define the signal-to-noise ratio $\rho_{i^*j} = \Delta_{i^*j}/\nu_{i^*j}$, and let $\rho_1 \stackrel{\text{def}}{=} \min_{j \neq i^*} \rho_{i^*j}$ be the minimum SNR over all rivals of i^*.

The rival $j^\dagger$ achieving ρ_1 is the hardest to reliably beat at position 1. Unlike Section A.6, $j^\dagger$ need not be the second-best model in expectation. To see why, suppose the second-best model j_1 has gap $\Delta_{i^*j_1} = 0.05$ and similar biases to i^*, giving $\nu_{i^*j_1} = \sigma\sqrt{2}$ and $\rho_{i^*j_1} = 0.05/(\sigma\sqrt{2})$. A weaker model j_2 has $\Delta_{i^*j_2} = 0.20$ but very different biases, giving $\nu_{i^*j_2} = 5\sigma\sqrt{2}$ and $\rho_{i^*j_2} = 0.04/(\sigma\sqrt{2}) < \rho_{i^*j_1}$. Despite being further behind on average, j_2 is harder to reliably beat because its performance relative to i^* swings widely across datasets.

Proposition A.6 (Position-1 disagreement under inductive biases)**.** *Under Assumptions A.1–A.5,*

$$\mathbb{P}[\varepsilon_1(N)] \;\leq\; 2(k-1)\,\Phi\Big(-\rho_1\sqrt{N}\Big).$$

Proof. Steps 1–2 are identical to Proposition A.5. In Step 3, Lemma A.2 gives $\bar{X}_{i^*}^{(B)} - \bar{X}_j^{(B)} \sim \mathcal{N}(\Delta_{i^*j}, \nu_{i^*j}^2/N)$, so standardising with standard deviation $\nu_{i^*j}/\sqrt{N}$ yields

$$\mathbb{P}\big[\bar{X}_j^{(B)} > \bar{X}_{i^*}^{(B)}\big] \;=\; \Phi(-\rho_{i^*j}\sqrt{N}).$$

Since $\rho_{i^*j} \geq \rho_1$ for all $j \neq i^*$ and Φ is increasing, $\Phi(-\rho_{i^*j}\sqrt{N}) \leq \Phi(-\rho_1\sqrt{N})$. Summing over $k-1$ rivals and applying Step 1 gives the result. □

Corollary A.6 (Asymptotic decay under inductive biases)**.** *As $N \to \infty$,*

$$\mathbb{P}[\varepsilon_1(N)] \;\leq\; \tilde{C}_1 \cdot N^{-1/2} \cdot \exp\left(-\frac{\rho_1^2 N}{2}\right), \qquad \tilde{C}_1 = \frac{2(k-1)}{\rho_1\sqrt{2\pi}}.$$

The decay exponent $-\rho_1^2/2$ reduces to $-\Delta_1^2/(4\sigma^2)$ of Corollary A.5 when $\nu_{i^*j^\dagger} = \sigma\sqrt{2}$ (no inductive bias), and is smaller whenever the inductive bias inflates $\nu_{i^*j^\dagger}$.

Comparison with Corollary A.3. Corollary A.3 showed that Kendall-τ disagreement under inductive biases decays with exponent $-\rho_{\min}^2/2$, where $\rho_{\min} = \min_{i<j} |\Delta_{ij}|/\nu_{ij}$ is the minimum SNR over all model pairs. Since ρ_1 is a minimum over only the pairs involving i^*, while $\rho_{\min}$ is a minimum over all pairs including those with no relation to i^*, we have $\rho_1 \geq \rho_{\min}$. This means the position-1 bound decays with a more negative exponent $-\rho_1^2/2 \leq -\rho_{\min}^2/2$, so fewer datasets are needed to reach any fixed error threshold ε. Setting each bound equal to ε and solving:

$$N^*_{\text{pos-1}} \;\propto\; \frac{2}{\rho_1^2}\log\frac{1}{\varepsilon}, \qquad N^*_{\text{Kendall}} \;\propto\; \frac{2}{\rho_{\min}^2}\log\frac{1}{\varepsilon}, \qquad \frac{N^*_{\text{pos-1}}}{N^*_{\text{Kendall}}} \;=\; \left(\frac{\rho_{\min}}{\rho_1}\right)^2 \leq 1.$$

Table B.1: Comparison of STRABLE against existing tabular benchmarks.

Benchmark	Attention to string features	Size and Scope
String-excluding benchmarks		
OpenML-CC18 [3]	Mostly numerical	High ($N = 72$); curated classification tasks.
PMLB / PMLBmini [53]	Numerical / Low-cardinality	High ($N \approx 290$); inclusive of simplified datasets.
Grinsztajn et al. (2022) [22]	One-Hot Encoding	Moderate ($N = 45$); removes high-cardinality features.
TabReD [54]	Removed / Numerical	Low ($N = 8$); industry-grade but removes string signals.
TabArena [12]	Curated / IID	Moderate ($N = 51$); does not include complex string signals.
String-flattening benchmarks		
McElfresh et al. (2024) [40]	Standard Vectorization	High ($N = 176$); relies on OpenML pre-processed formats.
AMLB [18]	AutoML System Specific Encoding	High ($N \approx 104$); focuses on AutoML framework evaluation.
TabRepo [55]	N-gram and Method Specific Encoding	High ($N = 211$); large-scale repository of model evaluations.
TALENT [68]	Standard Vectorization	High ($N = 300$); broad scope but standard numerical focus.
Zabërgja et al. (2025) [69]	Standard Vectorization	Moderate ($N = 68$); formally treats strings as mathematical vectors.
BeyondArena [47]	Fixed language model embeddings	High ($N = 142$); unified benchmark for diverse IID and non-IID tasks.
TEmBed [64]	Text Serialization	Moderate ($N = 69$); evaluates embeddings across cell, row, column, and table levels.
Narrow string-aware benchmarks		
Shi et al. (2021) [57]	Raw free-text	Low ($N = 18$); text-dominant tables with few tabular features.
CARTE [32]	LLM-embedded strings	Moderate ($N = 51$); curated datasets with discrete entries; lower density of text columns.
MulTaBench [2]	Target-aware fine-tuned encoders	Moderate ($N = 40$ core / 80 extended); isolates multimodal tasks where fine-tuned encoders improve performance, plus supplemental datasets.
TextTabBench [43]	Long free-text	Low ($N = 13$); limited dataset diversity.
STRABLE (Ours)	**Raw heterogeneous strings**	**High ($N = 108$); raw, uncurated, diverse string data.**

In other words, guaranteeing that two benchmarks agree on the top-ranked model requires fewer datasets than guaranteeing agreement on the full ranking. How many fewer depends on how much larger ρ_1 is than $\rho_{\min}$: if the hardest pair to separate is one involving i^* then $\rho_1 = \rho_{\min}$ and there is no advantage; if the hardest pair involves two mid-ranking models with nothing to do with i^* then $\rho_1 \gg \rho_{\min}$ and the advantage can be substantial.

Remark A.7 (Implications for STRABLE). Using Proposition A.5 alone, a practitioner would assess the difficulty of identifying Tf-Idf + TabPFN-2.5 as the top-1 model by looking at the mean gap Δ_1 to the second-best pipeline: large mean gap implies easy identification. Proposition A.6 asks a harder question: is that gap *consistent* across dataset types? In STRABLE, Section 5.3 shows that dataset characteristics such as string length and string diversity cause substantial shifts in the relative ordering of pipelines, meaning that the performance difference between the top pipeline and its rivals varies systematically with dataset type. This is a direct empirical signature of non-zero inductive bias terms $(\beta_{i^*} - \beta_j)^\top \Sigma_z (\beta_{i^*} - \beta_j)$, which inflate $\nu_{i^* j}$ and lower the SNR $\rho_{i^* j}$. Since $\rho_1 \leq \Delta_1/(\sigma\sqrt{2})$, Proposition A.6 requires at least as many datasets as Proposition A.5 to reach the same guarantee, and strictly more whenever any rival has inductive biases different from i^*.

Inductive biases introduce a new way for a rival model to be dangerous: a rival model may be far behind on average performance with respect to i^* but highly variable, therefore it can still frequently outscore i^* on specific benchmarks. Once accounted for that, one may need more datasets to wash out those accidental wins and stably identify i* as the winner.

B Current benchmark landscape

Table B.1 compares STRABLE against existing tabular benchmarks. Existing suites either focus on numerical data, rely on heavy curation to remove raw strings, or lack the scale of general-purpose benchmarks.

Table C.1: Tables per field and task type.

Field	b-cls	m-cls	reg	Total
Commerce	2	2	1	**5**
Economy	2	1	23	**26**
Education	1	0	9	**10**
Energy	0	0	9	**9**
Food	0	3	3	**6**
Health	8	9	13	**30**
Infra.	0	4	14	**18**
Social	0	0	4	**4**
Total	**13**	**19**	**76**	**108**

C Dataset collection

C.1 Dataset sources and characteristics

STRABLE is collected from 33 sources, spanning across 8 different domains (Table C.1). We targeted tables published by open-data providers (e.g., FDA, EMA, OSHA, World Bank), for three reasons. First, these providers cover the recurrent application areas of tabular ML — health, economic and business forecasting, education, energy, and infrastructure — which are also the domains represented in established benchmarks (e.g., TALENT spans finance, physics, and education). Second, their tables retain string columns rather than pre-encoded features, which is the setting STRABLE is designed to evaluate and which existing benchmarks exclude or flatten (Table B.1). Third, this string-heavy composition matches enterprise data: Kim et al. [34] report that enterprise tables are predominantly string-valued with low feature entropy due to repetitive values. Consistently, free text accounts for only 8.23% of STRABLE's string columns (Appendix C.4), while categorical, name and code entries dominate. Section 5.2 verifies empirically that pipeline rankings remain consistent across these domains.

Commerce (4 sources): European-Commission, webrobots.io, mercari.com, Yelp Open Dataset.

Economy (7 sources): aijobs.net, kaggle, Consumer-Financial-Protection-Bureau, Federal-Deposit-Insurance-Corporation, data.ct.gov, lendingclub.com, worldbankfinancesone.

Education (4 sources): commonlit.org, FSA, Institute of Museum and Library Services, SCIMAGO.

Energy (3 sources): energydata.info, fueleconomy.gov, world-resource-institute.

Food (6 sources): BeerAdvocate.com, flavorsofcacao.com, whiskyanalysis.com, Michelin, theramenrater.com, majestic.co.uk.

Health (6 sources): ClinicalTrials.gov, European-Medicines-Agency, FDA, HRSA, Medicaid, osha.gov.

Infrastructure (2 sources): HIFLD, data.sfgov.org.

Social (1 source): OHCA

In addition, Figure C.1 shows the datasets distribution per application field and year of assembling of the dataset (when unavailable, publication year was used); distribution of respective performances across all regression and classification tasks. Table C.2 shows the Median and inter-quartile range of different datasets features. In comparison, TextTaBench [43] is distinguished by significantly longer average string lengths, while CARTE [32] exhibits markedly higher cardinality distributions driven by categorical variations. STRABLE occupies a structural middle ground, featuring information-dense entries of moderate length and cardinality, distinct from both the long-context requirements of TextTaBench and the high-cardinality entity matching tasks of CARTE.

C.2 Detailed description of datasets

We provide detailed description and url links to the datasets. For broken links, refer to the link in the abstract to access the datasets.

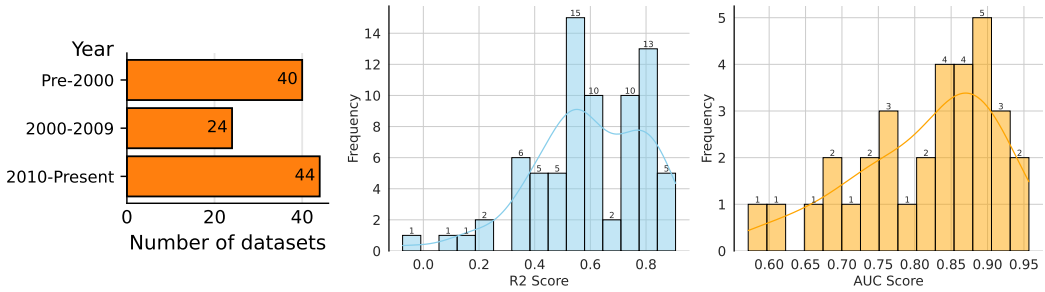


Figure C.1: From left to right: Number of datasets per publication or collection year, distribution of the R2 across all regression tasks, distribution of AUC across all classification tasks.

Table C.2: Summary statistics of curated datasets by category: Median [IQR]

Category	Number of Rows	Number of Columns	Number of String Columns	Avg. String Length	Cardinality
Commerce	75000 [42186, 75000]	14 [14, 19]	12 [9, 13]	49 [22, 58]	19158 [11476, 35516]
Economy	7796 [4723, 16160]	13 [11, 19]	10 [9, 17]	17 [12, 24]	1110 [288, 2567]
Education	10928 [5045, 21906]	16 [13, 29]	10 [8, 12]	16 [11, 26]	1894 [1518, 4850]
Energy	3978 [1238, 31448]	21 [19, 29]	17 [14, 23]	13 [12, 22]	416 [135, 3287]
Food	2993 [2056, 4554]	13 [8, 18]	8 [6, 13]	19 [10, 46]	888 [377, 1648]
Health	4366 [1743, 14988]	16 [10, 25]	12 [8, 22]	31 [14, 47]	696 [380, 1554]
Infrastructure	12504 [4530, 48014]	28 [24, 38]	17 [15, 21]	15 [13, 17]	2195 [1013, 5674]
Social	10081 [5637, 22476]	14 [12, 20]	12 [10, 16]	14 [12, 17]	490 [305, 672]

1. **ACA Federal Upper Limits**[5] Price limits for multi-source drugs under the Medicaid program. The task is to predict the federal upper price limit.
2. **AI/ML Salaries**[6] Salary and basic information for workers in the machine learning and data science industry. The task is to predict worker salaries.
3. **Animal and Veterinary Event**[7] Health problems reported in animals following the use of drug products. The task is to predict the severity of clinical signs.
4. **Antenna Structure Registration**[8] FCC registration data for antenna structures. The task is to predict the height of the structures.
5. **Awarded Grants IMLS**[9] Grants awarded by the Institute of Museum and Library Services. The task is to predict the specific grant amount.
6. **Beer Ratings**[10] Tasting profiles and consumer reviews for over 3,000 unique beers. The task is to predict overall review ratings.
7. **Broadband Availability**[11] Data on internet speed and availability across the US. The task is to predict the maximum available download speed.
8. **California Housing**[12] Median house values and demographics from the 1990 California census. The task is to predict median house prices.
9. **Child Adult Healthcare Quality**[13] Quality of care metrics for Medicaid and CHIP beneficiaries. The task is to predict healthcare performance scores.
10. **China Overseas Finance Inventory**[14] Chinese investments in power generation projects worldwide. The task is to predict the total investment amount.

[5] https://www.medicaid.gov/medicaid/prescription-drugs/federal-upper-limit
[6] https://ai-jobs.net/salaries/download/salaries.csv
[7] https://open.fda.gov/apis/animalandveterinary/event/
[8] https://hifld-geoplatform.opendata.arcgis.com/datasets/geoplatform::antenna-structure-registration/
[9] https://www.imls.gov/grants/awarded-grants
[10] https://www.kaggle.com/datasets/ruthgn/beer-profile-and-ratings-data-set
[11] https://www.fcc.gov/economics-analytics/broadband-data-collection
[12] https://www.kaggle.com/datasets/camnugent/california-housing-prices
[13] https://data.medicaid.gov/dataset/quality-of-care-child-core-set
[14] https://www.wri.org/data/china-overseas-finance-inventory-database

11. **Chocolate Bar Ratings**[15] Expert ratings and information on cocoa batches. The task is to predict the professional rating score.
12. **Clear Corpus**[16] Reading passage excerpts for elementary school students. The task is to predict the readability of the text.
13. **Cohort Default Rate**[17] Student loan default rates for US postsecondary institutions. The task is to predict the default rate percentage.
14. **College Credit Card Marketing**[18] Marketing agreements between credit card issuers and universities. The task is to predict the number of open accounts.
15. **College Deposit Product Marketing**[19] Agreements regarding deposit products offered to college students. The task is to predict associated financial metrics.
16. **Colleges and Universities**[20] Locations and characteristics of US postsecondary institutions. The task is to predict student enrollment.
17. **Commitments in Trust Funds**[21] Approved commitments in World Bank trust fund ledgers. The task is to predict the total commitment amount.
18. **Community Banking**[22] Financial metrics for community banks in the US. The task is to predict bank asset sizes or performance ratios.
19. **Conflict Events**[23] Geospatial data on political violence and protest events. The task is to predict the number of fatalities.
20. **Contract Awards IPF**[24] Contracts financed by the World Bank under Investment Project Financing. The task is to predict the award amount.
21. **Contributions to FIFs**[25] Financial contributions to multilateral Financial Intermediary Funds. The task is to predict donor contribution levels.
22. **Corporate Procurement Contracts**[26] Listing of contract awards executed by the World Bank Group. The task is to predict the contract value.
23. **Cosmetic Event**[27] Adverse events reported for cosmetic products to the FDA. The task is to predict the severity of the reported reaction.
24. **COVID-19 Clinical Trials**[28] Metadata for clinical trials related to COVID-19. The task is to predict trial status or enrollment numbers.
25. **Device Classification**[29] FDA classification of medical devices based on intended use. The task is to predict the specific device class.
26. **Device COVID-19 Serology**[30] Performance data for COVID-19 antibody tests. Each row is one test device applied to one serum sample from a fixed reference panel. The task is to predict whether the sample truly contains antibodies. Since every device is evaluated on the same reference samples, the label is almost fully determined by the pair (panel, sample number). Under row-level i.i.d. splits, this table behaves as a look-up task.
27. **Device PMA**[31] Premarket approval applications for high-risk medical devices. The task is to predict the final decision status.

[15] https://www.kaggle.com/datasets/rtatman/chocolate-bar-ratings
[16] https://www.commonlit.org/blog/introducing-the-clear-corpus-an-open-dataset-to-advance-research-28ff8cfea84a
[17] https://nsldsfap.ed.gov/cdr-searchable-database
[18] https://www.consumerfinance.gov/data-research/student-banking/marketing-agreements-and-data/
[19] https://www.consumerfinance.gov/data-research/student-banking/marketing-agreements-and-data/
[20] https://hifld-geoplatform.opendata.arcgis.com/datasets/geoplatform::colleges-and-universities/
[21] https://finances.worldbank.org/Trust-Funds-and-FIFs/Commitments-in-Trust-funds/DS00072
[22] https://www.fdic.gov/community-banking-research-program/reference-data
[23] https://data.humdata.org/dataset/acled-data
[24] https://finances.worldbank.org/Procurement/Contract-Awards-in-Investment-Project-Financing/DS00073
[25] https://finances.worldbank.org/Trust-Funds-and-FIFs/Contributions-to-Financial-Intermediary-Funds/DS00051
[26] https://finances.worldbank.org/Procurement/Corporate-Procurement-Contract-Awards/DS00028
[27] https://open.fda.gov/apis/cosmetic/event/
[28] https://clinicaltrials.gov/ct2/results?cond=COVID-19
[29] https://open.fda.gov/apis/device/classification/
[30] https://open.fda.gov/apis/device/covid19serology/
[31] https://open.fda.gov/apis/device/pma/

28. **Disbursements in Trust Funds**[32] Cash payments made to recipients of World Bank trust funds. The task is to predict the disbursement amount.
29. **Discretionary Grant**[33] Grants awarded based on competitive applications by HRSA. The task is to predict the total grant award amount.
30. **Drug Drugs@FDA**[34] Information about FDA-approved brand name and generic drugs. The task is to predict approval years.
31. **Drug Enforcement**[35] FDA enforcement actions related to drug products. The task is to predict the recall classification level.
32. **Drug NDC**[36] National Drug Code Directory containing all drugs in the US. The task is to predict the drug category.
33. **Drug Shortages**[37] Information on current and resolved drug shortages. The task is to predict the duration of the shortage.
34. **Electric Generating Plants**[38] Operational characteristics of power plants in the US. The task is to predict net generation capacity.
35. **Electric Retail Service Territories**[39] Areas served by electric utility companies. The task is to predict the utility ownership type.
36. **EMA Medicines**[40] Information on medicines authorized by the European Medicines Agency. The task is to predict authorization status.
37. **External Clinician Dashboard**[41] Performance metrics for clinicians in HRSA programs. The task is to predict clinician productivity scores.
38. **FIF Cash Transfers**[42] Cash transfers from FIFs to implementing agencies. The task is to predict the transfer amount.
39. **FIF Commitments**[43] Funding commitments made by Financial Intermediary Funds. The task is to predict the commitment value.
40. **FIF Funding Decisions**[44] Decisions on funding allocations by FIF governing bodies. The task is to predict the decision outcome.
41. **Financial Management Medicaid**[45] State expenditures on Medicaid programs and services. The task is to predict total program costs.
42. **Financial Product Complaint**[46] Consumer complaints regarding financial products and services. The task is to predict the response category.
43. **Food Enforcement**[47] Recall and enforcement actions for food products. The task is to predict the reason for the recall.
44. **Food Event**[48] Adverse events and product complaints for food and supplements. The task is to predict the consumer outcome.
45. **Food Prices**[49] Historical food prices from markets worldwide. The task is to predict the price of staple crops.

[32] https://finances.worldbank.org/Trust-Funds-and-FIFs/Disbursements-in-Trust-Funds/DS00074
[33] https://data.hrsa.gov/data/reports
[34] https://open.fda.gov/apis/drug/drugsfda/
[35] https://open.fda.gov/apis/drug/enforcement/
[36] https://open.fda.gov/apis/drug/ndc/
[37] https://www.fda.gov/drugs/drug-safety-and-availability/drug-shortages
[38] https://hifld-geoplatform.opendata.arcgis.com/datasets/geoplatform::power-plants/
[39] https://hifld-geoplatform.opendata.arcgis.com/datasets/geoplatform::electric-retail-service-territories/
[40] https://www.ema.europa.eu/en/medicines/download-medicine-data
[41] https://data.hrsa.gov/data/reports
[42] https://finances.worldbank.org/Trust-Funds-and-FIFs/Financial-Intermediary-Funds-Cash-Transfers/DS00029
[43] https://finances.worldbank.org/Trust-Funds-and-FIFs/Financial-Intermediary-Funds-Commitments/DS00052
[44] https://finances.worldbank.org/Trust-Funds-and-FIFs/Financial-Intermediary-Funds-Funding-Decisions/DS00053
[45] https://data.medicaid.gov/dataset/state-expenditures
[46] https://www.consumerfinance.gov/data-research/consumer-complaints/
[47] https://open.fda.gov/apis/food/enforcement/
[48] https://open.fda.gov/apis/food/event/
[49] https://data.humdata.org/dataset/wfp-food-prices

46. **Foreign Gift and Contract**[50] Reports of gifts or contracts from foreign sources to US colleges. The task is to predict the gift value.
47. **FTS Funding**[51] Humanitarian aid flows tracked by the Financial Tracking Service. The task is to predict funding per crisis.
48. **FTS Requirements and Funding**[52] Requirements vs. funding for humanitarian response plans. The task is to predict the funding gap.
49. **Gainful Employment**[53] Debt-to-earnings ratios for graduates of vocational programs. The task is to predict if a program passes standards.
50. **Global Dams Database**[54] Geographic and structural information on dams worldwide. The task is to predict dam capacity.
51. **Global Power Plant**[55] Comprehensive database of power plants worldwide. The task is to predict annual electricity generation.
52. **Grant**[56] General information on federal grants awarded by HHS. The task is to predict the project funding amount.
53. **Health Professional Shortage Areas**[57] Regions with a shortage of healthcare providers. The task is to predict the shortage score.
54. **Historic Perimeters Wildfires**[58] Historical boundaries of major wildfires in the US. The task is to predict total acres burned.
55. **Historical Earthquake Locations**[59] Global database of significant historical earthquakes. The task is to predict earthquake magnitude.
56. **Historical Volcanic Locations**[60] Locations and eruption history of significant volcanoes. The task is to predict the eruption type.
57. **Hospitals**[61] Comprehensive list of US hospitals and their facilities. The task is to predict the number of hospital beds.
58. **Hypertension Control**[62] Clinical performance data on hypertension management. The task is to predict patient blood pressure control percentage.
59. **IBRD Statement of Loans**[63] Historical record of loans and guarantees issued by the IBRD. The task is to predict the loan status.
60. **IDA Statement of Credits**[64] Records of credits, grants, and guarantees issued by IDA. The task is to predict the disbursement status.
61. **IFC Advisory Projects**[65] Metadata on advisory services provided by the IFC. The task is to predict the project budget.
62. **IFC Investment Projects**[66] Records of investment projects undertaken by the IFC. The task is to predict the investment amount.

[50] https://studentaid.gov/data-center/school/foreign-gifts
[51] https://fts.unocha.org/
[52] https://fts.unocha.org/
[53] https://studentaid.gov/data-center/school/ge
[54] https://www.energydata.info/dataset/global-dams-database
[55] https://www.wri.org/data/global-power-plant-database
[56] https://taggs.hhs.gov/
[57] https://data.hrsa.gov/topics/health-workforce/shortage-areas
[58] https://hifld-geoplatform.opendata.arcgis.com/datasets/geoplatform::historic-perimeters-wildfires/
[59] https://hifld-geoplatform.opendata.arcgis.com/datasets/geoplatform::historical-earthquake-locations/
[60] https://hifld-geoplatform.opendata.arcgis.com/datasets/geoplatform::historical-volcanic-locations/
[61] https://hifld-geoplatform.opendata.arcgis.com/datasets/geoplatform::hospitals/
[62] https://data.hrsa.gov/data/reports
[63] https://finances.worldbank.org/Loans-and-Credits/IBRD-Statement-of-Loans-Latest-Available-Snapshot/DS00049
[64] https://finances.worldbank.org/Loans-and-Credits/IDA-Statement-of-Credits-and-Grants-Latest-Availabl/DS00048
[65] https://finances.worldbank.org/Advisory-Services/IFC-Advisory-Services-Projects/DS00010
[66] https://finances.worldbank.org/Investment-Services/IFC-Investment-Services-Projects/DS00011

63. **Industry Payments Entity**[67] Payments made by drug and device companies to teaching hospitals. The task is to predict the payment amount.
64. **Industry Payments Project**[68] Payments related to specific research projects or clinical trials. The task is to predict research funding.
65. **Insurance Company Complaints**[69] Consumer complaints filed against insurance companies. The task is to predict resolution status.
66. **Journal Ranking**[70] Scientific journal metrics including H-index and citations. The task is to predict the journal's impact factor.
67. **Kickstarter Projects**[71] Funding goals and outcomes for Kickstarter campaigns. The task is to predict project success.
68. **Lending Club Loan**[72] Information on loans issued through the Lending Club platform. The task is to predict the interest rate.
69. **Local Government Renewable Action**[73] Renewable energy initiatives taken by local governments. The task is to predict project capacity.
70. **Local Law Enforcement**[74] Locations of local police and sheriff departments in the US. The task is to predict officer counts.
71. **Managed Care Enrollment**[75] Enrollment statistics for Medicaid managed care plans. The task is to predict the number of enrollees.
72. **Media Ranking**[76] Rankings for media and social science publications. The task is to predict the impact factor.
73. **Medically Underserved Areas**[77] Areas with populations lacking access to primary care. The task is to predict the underservice index.
74. **Mercari Price Prediction**[78] Product descriptions and categories from Mercari. The task is to predict the listing price.
75. **Michelin Ratings**[79] Details on restaurants curated in the Michelin Guide. The task is to predict the award level.
76. **MIGA Issued Projects**[80] Projects supported by MIGA investment guarantees. The task is to predict the maximum gross exposure.
77. **MLR Summary Reports**[81] Medical Loss Ratio data for healthcare plans. The task is to predict the MLR percentage.
78. **Mobile Home Parks**[82] Geographic locations and capacities of mobile home parks. The task is to predict the number of lots.
79. **Museums**[83] Information on museums and related organizations in the US. The task is to predict annual revenue.
80. **NADAC Rates**[84] Weekly survey of drug acquisition costs for retail pharmacies. The task is to predict the acquisition cost per unit.
81. **National Average Drug Acquisition Cost**[85] Survey of retail pharmacy drug acquisition costs. The task is to predict units costs.

[67] https://www.cms.gov/openpayments
[68] https://www.cms.gov/openpayments
[69] https://data.ct.gov/Business/Insurance-Company-Complaints/
[70] https://www.scimagojr.com/journalrank.php
[71] https://www.kaggle.com/datasets/kemical/kickstarter-projects
[72] https://www.lendingclub.com/info/download-data.action
[73] https://www.wri.org/data/local-government-renewables-action-tracker
[74] https://hifld-geoplatform.opendata.arcgis.com/datasets/geoplatform::local-law-enforcement-locations/
[75] https://data.medicaid.gov/dataset/managed-care-enrollment-report
[76] https://www.scimagojr.com/
[77] https://data.hrsa.gov/topics/health-workforce/shortage-areas
[78] https://www.kaggle.com/c/mercari-price-suggestion-challenge
[79] https://www.kaggle.com/datasets/ngshiheng/michelin-guide-restaurants-2021
[80] https://finances.worldbank.org/Guarantees/MIGA-Issued-Projects/DS00030
[81] https://data.medicaid.gov/dataset/medical-loss-ratio
[82] https://hifld-geoplatform.opendata.arcgis.com/datasets/geoplatform::mobile-home-parks/
[83] https://www.imls.gov/research-evaluation/data-collection/museum-universe-data-file
[84] https://data.medicaid.gov/dataset/national-average-drug-acquisition-cost
[85] https://data.medicaid.gov/dataset/national-average-drug-acquisition-cost

82. **Oil and Natural Gas Platforms**[86] Offshore oil and gas platforms in US waters. The task is to predict platform status.
83. **Orphan Designations**[87] Medicines designated for rare diseases by the EMA. The task is to predict therapeutic area.
84. **OSHA Accidents**[88] Reports of workplace accidents and fatalities. The task is to predict injury severity.
85. **Paediatric Investigation Plan**[89] Research plans for the use of medicines in children. The task is to predict investigation status.
86. **POL Terminal**[90] Petroleum, Oil, and Lubricant storage terminals. The task is to predict storage capacity.
87. **Power Plants**[91] Details on fuel type and location of US power plants. The task is to predict the energy source.
88. **Prepaid Financial Product**[92] Information on prepaid financial products and terms. The task is to predict fee structures.
89. **Prison Boundaries**[93] Geospatial boundaries and capacities of US correctional facilities. The task is to predict population.
90. **Ramen Ratings**[94] Reviews and ratings for various ramen products globally. The task is to predict the star rating.
91. **RASFF Window**[95] Food and feed safety alerts from the EU Rapid Alert System. The task is to predict the risk level.
92. **RASNF Notification List**[96] The EU rapid alert system for dangerous non-food products. The task is to predict whether the product affects more than one country.
93. **Recipient Executed Grants**[97] Commitments and disbursements for grants executed by recipients. The task is to predict grant value.
94. **Schools**[98] Locations and metadata for public K-12 schools. The task is to predict student counts.
95. **SF Building Permits**[99] Building permit applications in San Francisco. The task is to predict construction cost.
96. **Summary of Deposit**[100] Branch-level deposit data for US banks. The task is to predict total deposits per branch.
97. **Tax Incentives**[101] Business tax incentives granted by Connecticut. The task is to predict the tax credit amount.
98. **Terms CC Plans**[102] Terms and conditions for various credit card plans. The task is to predict interest rates.
99. **Tobacco Problem**[103] Health or product problems related to tobacco. The task is to predict the health problem type.

[86] https://hifld-geoplatform.opendata.arcgis.com/datasets/geoplatform::oil-natural-gas-platforms/
[87] https://www.ema.europa.eu/en/medicines/download-medicine-data
[88] https://www.osha.gov/fatalities
[89] https://www.ema.europa.eu/en/medicines/download-medicine-data
[90] https://hifld-geoplatform.opendata.arcgis.com/datasets/geoplatform::petroleum-terminals/
[91] https://hifld-geoplatform.opendata.arcgis.com/datasets/geoplatform::power-plants/
[92] https://www.consumerfinance.gov/data-research/prepaid-accounts/
[93] https://hifld-geoplatform.opendata.arcgis.com/datasets/geoplatform::prison-boundaries/
[94] https://www.kaggle.com/datasets/ankanhore545/top-ramen-ratings-2022
[95] https://ec.europa.eu/food/safety/rasff_en
[96] https://ec.europa.eu/safety-gate-alerts/screen/search
[97] https://finances.worldbank.org/Trust-Funds-and-FIFs/Recipient-executed-grants-commitments-disbursement/DS00075
[98] https://hifld-geoplatform.opendata.arcgis.com/datasets/geoplatform::public-schools/
[99] https://data.sfgov.org/Housing-and-Buildings/Building-Permits/i98e-oved
[100] https://www.fdic.gov/bank/statistical/stats/
[101] https://data.ct.gov/Business/Tax-Incentives/
[102] https://www.consumerfinance.gov/data-research/credit-card-data/
[103] https://open.fda.gov/apis/tobacco/problem/

100. **Total Contributions IBRD IDA IFC**[104] Contributions to World Bank institutions by members. The task is to predict total contribution amounts.
101. **Transmission Lines**[105] Electric power transmission infrastructure in the US. The task is to predict voltage levels.
102. **Transmission Towers**[106] Structures for wireless and broadcast transmission. The task is to predict tower height.
103. **US School Bus Fleet**[107] Data on school bus fleets and fuel types in the US. The task is to predict bus counts.
104. **Vehicles**[108] Fuel economy data for cars sold in the US. The task is to predict the annual fuel cost.
105. **Whisky Ratings**[109] Tasting notes and meta-critic scores for whiskies. The task is to predict the overall rating.
106. **Wine Dataset**[110] Wine reviews and prices from Wine Enthusiast magazine. The task is to predict the score or price.
107. **Workforce Demographics**[111] Demographic information for the health professional workforce. The task is to predict regional workforce density.
108. **Yelp Business**[112] Metadata on businesses including categories and reviews. The task is to predict the star rating.

C.3 Downsampling Strategy

To ensure computational feasibility across our extensive benchmark of learners and encoders, we limit the maximum number of data points to $75{,}000$. For datasets exceeding the limit, they are downsampled using a fixed random seed for reproducibility. The sampling strategy depends on the task type: simple random sampling (uniform selection without replacement) for regression; stratified sampling to ensure that the class distribution in the subset matches the original target marginal distribution for classification. The threshold of $75{,}000$ is chosen to align with TabPFN-2.5's design specification for datasets of up to $50{,}000$ training samples [24]: under our 3-fold cross-validation, this threshold ensures each training fold contains at most $50{,}000$ samples ($\approx 2/3 \times 75{,}000$), with the remaining $25{,}000$ used for testing.

While the value is set by TabPFN-2.5, the cap itself is applied identically to all pipelines: our rank comparisons and Friedman/Conover–Iman tests are paired designs requiring the same data across pipelines, so capping only input-limited models would confound model quality with training-set size. Evaluating at fixed training sizes is standard practice in tabular benchmarks [22, 32], and a cap is needed regardless to bound the cost of 445 pipelines on tables with millions of rows (e.g., `financial-product-complaint`, 4.86M).

The cap affects 12 of the 108 tables, and we verified that it does not drive our conclusions. First, running 8 of these 12 tables at full size preserves the pipeline ranking (Kendall-$\tau = 0.96$, Figure E.11). Second, ranking pipelines on the 96 uncapped tables alone reproduces the global ranking ($\tau = 0.96$) with the same top-5 pipelines. The leading position of Tf-Idf + TabPFN-2.5 is therefore not an artifact of the cap.

C.4 String Taxonomy and Profiling Methodology

To better characterize the string modalities present within the STRABLE benchmark, we introduced a semantic taxonomy and applied it to profile all text columns across the curated datasets.

[104] https://finances.worldbank.org/
[105] https://hifld-geoplatform.opendata.arcgis.com/datasets/geoplatform::electric-power-transmission-lines/
[106] https://hifld-geoplatform.opendata.arcgis.com/datasets/geoplatform::antenna-structure-registration/
[107] https://hifld-geoplatform.opendata.arcgis.com/datasets/geoplatform::school-buses/
[108] https://www.fueleconomy.gov/feg/download.shtml
[109] https://whiskyanalysis.com/index.php/database/
[110] https://www.kaggle.com/datasets/manyregression/updated-wine-enthusiast-review
[111] https://data.hrsa.gov/data/reports
[112] https://www.yelp.com/dataset

Table C.3: Median (IQR) of Runtime and MTEB Performance per Language Model

Language Model	Hugging Face	Median (IQR) Runtime [s]	MTEB (En) Score
LM All-MiniLM-L12-v2	sentence-transformers/all-MiniLM-L12-v2	12 [4, 48]	-
LM All-MiniLM-L6-v2	sentence-transformers/all-MiniLM-L6-v2	3 [1, 12]	56.03
LM All-MPNet-base-v2	sentence-transformers/all-mpnet-base-v2	13 [4, 55]	-
LM BGE-base	BAAI/bge-base-en-v1.5	11 [4, 32]	65.14
LM BGE-large	BAAI/bge-large-en-v1.5	35 [9, 100]	65.89
LM BGE-small	BAAI/bge-small-en-v1.5	12 [3, 42]	64.30
LM DeBERTa-v3-base	microsoft/deberta-v3-base	28 [10, 103]	-
LM DeBERTa-v3-large	microsoft/deberta-v3-large	48 [14, 182]	-
LM DeBERTa-v3-small	microsoft/deberta-v3-small	12 [6, 35]	-
LM DeBERTa-v3-xsmall	microsoft/deberta-v3-xsmall	20 [6, 75]	-
LM E5-base-v2	intfloat/e5-base-v2	7 [2, 24]	61.67
LM E5-large-v2	intfloat/e5-large-v2	16 [4, 59]	62.79
LM E5-small-v2	intfloat/e5-small-v2	6 [2, 19]	61.32
LM F2LLM-0.6B	codefuse-ai/F2LLM-0.6B	52 [14, 191]	70.03
LM F2LLM-1.7B	codefuse-ai/F2LLM-1.7B	72 [17, 279]	72.01
LM F2LLM-4B	codefuse-ai/F2LLM-4B	290 [74, 997]	73.67
LM FastText	-	0 [0, 1]	-
LM Gemma-0.3B	google/gemma-3-270m	47 [14, 157]	-
LM Jasper-0.6B	infgrad/Jasper-Token-Compression-600M	22 [7, 77]	74.75
LM KALM-embed	HIT-TMG/KaLM-embedding-multilingual-mini-instruct-v1.5	56 [15, 210]	71.29
LM LLaMA-3.1-8B	meta-llama/Llama-3.1-8B	249 [61, 1040]	-
LM LLaMA-3.2-1B	meta-llama/Llama-3.2-1B	36 [9, 155]	-
LM LLaMA-3.2-3B	meta-llama/Llama-3.2-3B	106 [26, 439]	-
LM LLaMA-Nemotron-Embed-1B-v2	nvidia/llama-nemotron-embed-1b-v2	35 [8, 142]	-
LM ModernBERT-base	answerdotai/ModernBERT-base	26 [8, 93]	-
LM ModernBERT-large	answerdotai/ModernBERT-large	35 [11, 119]	-
LM OPT-0.1B	facebook/opt-125m	13 [3, 45]	-
LM OPT-0.3B	facebook/opt-350m	25 [6, 90]	-
LM OPT-1.3B	facebook/opt-1.3b	57 [14, 219]	-
LM OPT-2.7B	facebook/opt-2.7b	101 [24, 437]	-
LM OPT-6.7B	facebook/opt-6.7b	233 [58, 939]	-
LM Qwen-3-0.6B	Qwen/Qwen3-Embedding-0.6B	30 [9, 140]	70.47
LM Qwen-3-4B	Qwen/Qwen3-Embedding-4B	156 [40, 653]	74.61
LM Qwen-3-8B	Qwen/Qwen3-Embedding-8B	276 [72, 1174]	75.23
LM RoBERTa-base	FacebookAI/roberta-base	7 [2, 25]	-
LM RoBERTa-large	FacebookAI/roberta-large	20 [5, 73]	-
LM Sentence-T5-base	sentence-transformers/sentence-t5-base	8 [3, 36]	60.30
LM Sentence-T5-large	sentence-transformers/sentence-t5-large	17 [4, 62]	77.67
LM Sentence-T5-xl	sentence-transformers/sentence-t5-xl	61 [14, 259]	76.58
LM Sentence-T5-XXL	sentence-transformers/sentence-t5-xxl	268 [65, 1019]	66.13
LM UAE-large	WhereIsAI/UAE-Large-V1	24 [6, 99]	66.40

Semantic Taxonomy We categorize all non-numeric columns into six distinct semantic types:

- **Categorical:** Low-uniqueness, repeating labels (e.g., "Red", "General Acute Care").
- **Name:** Proper nouns representing people, organizations, places, or products (e.g., "John Doe", "Max Mara").
- **Structured Code:** Strings with recognizable, meaningful patterns (e.g., ZIP codes, ICD/NDC medical codes, URLs).
- **Free Text:** Multi-word prose containing natural language and stopwords (e.g., user reviews, medical notes).
- **Identifier:** Near-unique, opaque keys with no inherent semantics (e.g., UUIDs, hashes, auto-generated IDs).
- **Datetime:** Strings encoding temporal information (e.g., "2024-03-15", "March 15, 2024", "Q1 2024").

Profiling Methodology To classify these columns at scale, we first isolate all text columns using `skrub.TableVectorizer`. We then compute a suite of deterministic indices for each column, which include:

- *Dictionary Hit Rate & Stopword Density:* To distinguish natural language prose from random strings.
- *Symbol Density, Proportion Numeric, & Pattern Matching:* To detect structured formatting (such as slashes and dashes), numeric content, and standard regex patterns (such as dates and currencies).
- *Token Metrics:* Average words per cell and the proportion of multi-word entries.
- *Uniqueness Ratio:* To differentiate primary keys from repeating categorical variables.

We apply a heuristic function based on these computed indices to classify each column into one of the six taxonomic tags.

Validation Protocol To ensure the accuracy of our heuristic profiler, we conducted a rigorous validation loop on a sample of 30 datasets. We manually annotated the semantic types of the string columns and compared them against the heuristic's output, utilizing a State-of-the-Art LLM-as-a-judge as a secondary verifier. By iteratively refining the heuristic thresholds based on failure cases in this sample, our automated profiler achieved a 97% agreement with the ground truth annotations.

D Evaluation pipeline

D.1 Pipeline components

To support the evaluation of the STRABLE benchmark, we use a comprehensive corpus of encoding and learning components. These are categorized into modular encoder-learner pipelines and end-to-end models.

Encoder-learner pipelines We consider combinations of following encoders and learners.

Baselines of encoders:

- `Tf-Idf+SVD`: encodes strings of a given column using tf-idf vectorization and truncated singular value decomposition for dimensionality reduction. We use the `Skrub` package [58] with its default value 30 as the dimension.
- `TargetEncoder` [41]: encodes categorical variables based on the global target mean and target values for observations belonging to the category. Categories that are not present in the train set are encoded with the target mean.
- `LM-`: a family of encoders that use pre-trained language models as feature extractors. For models on `HuggingFace`, we use the `Sentence-Transformers` package to extract the embeddings; for `FastText`, we rely on its supported package. The encoding is then followed by Principal Component Analysis (PCA) to reduce each column's representation to 30 components. PCA is fitted with only non-null values, preserving missing entries as NaNs.
- `TARTE` [33]: a model pre-trained from large knowledgebases. The model takes a table as the input and generates embeddings per row, with dimension of 768.

Baselines of learners:

- `Ridge` [45]: simple linear model with an efficient cross-validation for hyperparameter selection. We record the results of internally tuned estimator.
- `XGBoost` [6]: a representative gradient boosting decision trees learner. Models are configured with a maximum of $1,000$ iterations with early-stopping (patience=300) based on a validation set. We record results of both default and tuned estimators.
- `ExtraTrees` [17]: fits a number of randomized decision trees on various sub-samples of the dataset with averaging for improved prediction. Implemented via `scikit-learn==1.7.2`, which provides native missing value support. We record results of both default and tuned estimators.

- `RealMLP` [29]: an improved MLP with architectural changes and meta-tuned default hyperparameters, specifically optimized for tabular data. We use the default configuration.
- `TabM` [19]: a tabular deep learning model based on parameter-efficient ensembling of MLPs via BatchEnsemble, producing multiple predictions per object. We use the default configuration.
- `TabPFN-2.5` [24]: in-context learner leveraging prior-data fitted networks trained with synthetic data. We use the official `2.5` release, with the default configurations ("TabPFN-2.5" and "Real-TabPFN-2.5" for regression and classification, respectively).
- `TabICLv2` [49]: a tabular foundation model for in-context learning that extends TabICL with a novel synthetic-data engine, a scalable softmax for longer contexts, and improved pretraining. We use the default configuration. TabICLv2 was pretrained on up to 100 features, which is below the typical post-PCA feature count of our benchmark (mean 416, max 1270 features after 30-PCA per high-cardinality column; see Table E.11); this likely contributes to its slightly lower performance compared to TabPFN-2.5, which was pretrained on up to 2000 features (Figure 4).

End-to-end models End-to-end models jointly process encoders and learners for a given table (a generative LLM predictor, `TabuLa-8B`, is evaluated separately on five small tables in Table E.6):

- `ConTextTab` [59]: in-context learner that has been pre-trained from the real-world T4 dataset curated in Gardner et al. [16]. The learner combines string encodings from `All-MiniLM-L6-v2` with cross-row and cross-column attention [36]. The model is instantiated from the `SAP_RPT_OSS` package using model defaults.
- `TabSTAR` [1]: a pre-trained model from real world datasets with rich semantic information. The architecture integrates `e5-small-v2` that is tuned alongside the tabular backbone. The model is evaluated with the default parameters.
- `CatBoost` [46]: a gradient-boosted trees package commonly used to learn on tables. Uses internal handling of categorical attributes and receives categorical feature indices explicitly from the encoding pipeline. We treat text features as categorical, encoded by CatBoost's categorical encoding, an improved version of target encoding. Models are configured with a maximum of $1,000$ iterations with early-stopping with `od_type='Iter'` (patience=300) based on a validation set. We record results of both default and tuned estimators.
- `Mambular` [61]: a tabular deep-learning architecture that treats features as a pseudo-sequence and processes them through Mamba state-space blocks. We use the `MambularClassifier` and `MambularRegressor` interfaces from `deeptab` with default parameters.

Note on `ConTextTab` pretraining-contamination risk. `ConTextTab` [59] is pretrained on 2.18M tables from the T4 corpus [16], a Common Crawl and GitHub snapshot whose provenance has been shown to inflate `TabuLa-8B`'s results via train–test overlap [20]. We therefore ran a leakage-detection procedure designed to be as strict as possible, i.e., to flag as many T4 tables as plausibly overlap with STRABLE.

Every cell is canonicalized so that formatting differences cannot hide a match: lowercasing, whitespace stripping, number normalization (`3.140` $\equiv$ `3.14`) and null removal, followed by hashing to `uint64`. The canonicalized STRABLE tables are indexed in memory through three structures:

- **Distinctive-value index.** Highly specific text strings (non-numeric, ≥ 5 characters), to flag tables sharing rare identifiers.
- **Row-fingerprint index.** A column-order-independent hash of every row, as a fallback for low-entropy tables containing only common numbers or short codes.
- **Per-column value sets.** The unique values of every STRABLE column, to compute common-value ratios between T4 and STRABLE columns.

The entire T4 corpus, deduplicated by content hashing to 2.89M unique tables, is processed in a single sequential pass. The vast majority (99.99%) of T4 tables share no rare value with STRABLE and are discarded by dictionary lookup; for the remainder we retain the top-K highest-overlapping candidates per STRABLE table.

To separate genuine leaks from coincidental overlap, a value is considered *rare* if it appears in ≤ 100 of the 2.89M unique tables ($\approx 0.003\%$ of the corpus), which removes false positives from

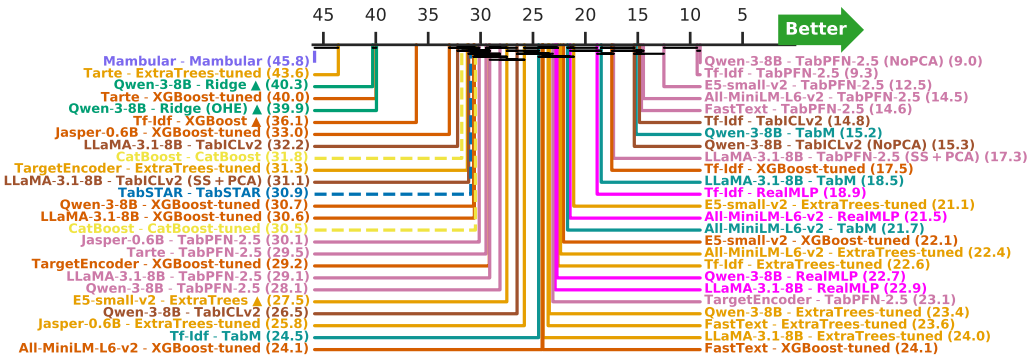


Figure D.1: **Critical difference diagram of Figure 3 recomputed without ConTextTab.** Removing ConTextTab leaves the ranking unchanged (Kendall $\tau = 0.998$ with the order of Figure 3). The only change is a swap between two near-tied mid-ranking pipelines, LLaMA-3.1-8B + ExtraTrees-tuned and FastText + XGBoost-tuned (24th/25th, mean ranks within 0.01 of each other); in particular, the top-10 and bottom-5 pipelines keep exactly the same order. Qwen-3-8B + TabPFN-2.5 (NoPCA) remains in first place (mean rank 9.3 $\rightarrow$ 9.0), followed by Tf-Idf + TabPFN-2.5 (9.6 $\rightarrow$ 9.3).

shared generic vocabulary (state names, region codes). Containment is measured column-wise as $|S_{\text{col}} \cap T_{\text{col}}|/|S_{\text{col}}|$, with S and T the STRABLE and T4 columns; the asymmetry lets us catch T4 tables holding only a row-subset of the original.

A T4 table is flagged as contaminated with respect to a STRABLE table if it satisfies any of:

- **Structural rule:** ≥ 2 shared rare values and either ≥ 3 columns with containment > 0.7, or mean column containment > 0.4. Being content-based, it is robust to column renaming.
- **Schema rule:** ≥ 4 matching column names covering $\geq 50\%$ of the STRABLE table's columns, together with ≥ 3 shared rare values.
- **Content rule:** ≥ 8 shared rare values globally. This threshold is intentionally loose, to capture partial or transformed duplicates.

Under these rules, 2% of the T4 corpus exhibits plausible overlap with STRABLE. To check whether `ConTextTab`'s inclusion distorts the comparison of the other pipelines, we re-run the aggregate analysis with `ConTextTab` removed from the pool (Figure D.1): the relative standings, pairwise performance gaps and convergence trends of all other pipelines are unchanged.

Cross-validation protocol To evaluate STRABLE, we employ a nested cross-validation protocol consisting of outer loop for performance measurement and inner loop for hyperparameter selection. For the outer loop, a dataset is partitioned into 3 folds, with additional stratified sampling for classification tasks. In each iteration, one fold is held out as the test set, while the remaining folds constitute the training partition. Within each outer training partition, we perform hyperparameter selection via an internal 8-fold cross-validation as done in Erickson et al. [12].

Hyperparameter optimization and prediction For baseline estimators requiring hyperparameter optimization, we conduct a randomized search over 100 iterations (including the default values). The configuration achieving the highest mean validation score in the inner loop is selected. Detailed search spaces can be found in Table D.1. For final prediction, we employ two schemes: For learners that use a validation-set (*e.g.,* `XGBoost`), we average the predictions of the 8 folds, following Erickson et al. [12]; otherwise, we refit a model using the total training partition.

Evaluation metrics We report predictive performance and computational efficiency, with metrics of predictive power (R^2 for regression; AUROC for classification), durations of preprocessing and hyperparameter search, and inference latency. For regression tasks, we apply parameter-free target transformations (log, log1p, cbrt, arcsinh, signed-log) selected per dataset at preprocessing time using a skewness-minimization criterion; R^2 is reported on the transformed scale. We verify that

this choice does not affect model rankings between raw and transformed targets across 61 regression tasks (Kendall-$\tau = 0.95$, Figure E.9).

Statistical significance tests To assess the statistical significance of the performance differences between the k evaluated pipelines across N datasets, we first employ the **Friedman test** [14], a non-parametric equivalent of ANOVA for repeated measures. The null hypothesis states that all algorithms perform equivalently in terms of average rank.

Upon rejection of the null hypothesis ($p < 0.05$), we employ the **Conover-Iman post-hoc test** [7] for pairwise comparisons. The test statistic for comparing two algorithms i and j is given by:

$$T = \frac{|\bar{R}_i - \bar{R}_j|}{\sqrt{\hat{S}^2 \left(\frac{2}{N}\right)}} \tag{D.1}$$

where $\bar{R}_i$ and $\bar{R}_j$ are the average ranks of the algorithms, and $\hat{S}^2$ is the pooled sample variance of the ranks. The difference between two algorithms is considered statistically significant if the p-value derived from the t-distribution is below the significance level $\alpha = 0.05$.

Table D.1: Hyperparameter search space for STRABLE learners.

Methods	Parameters	Grid
TabPFN-2.5	-	Default parameters
TabStar	-	Default parameters
ConTextTab	-	Default parameters
Ridge Regression	Alpha (α)	$[0.01, 0.1, 1, 10, 100]$
XGBoost	Max depth	UniformInt $[2, 6]$
	Min child weight	LogUniform $[1, 100]$
	Subsample	Uniform $[0.5, 1]$
	Learning rate	LogUniform $[10^{-5}, 1]$
	Colsample by level	Uniform $[0.5, 1]$
	Colsample by tree	Uniform $[0.5, 1]$
	Gamma	LogUniform $[10^{-8}, 7]$
	L2 regularization (λ)	LogUniform $[1, 4]$
	L1 regularization (α)	LogUniform $[10^{-8}, 100]$
CatBoost	Max depth	UniformInt $[2, 6]$
	Learning rate	LogUniform $[10^{-5}, 1]$
	Bagging temperature	Uniform $[0, 1]$
	l_2-leaf regularization	LogUniform $[1, 10]$
	Random strength	UniformInt $[1, 20]$
	One hot max size	UniformInt $[0, 25]$
	Leaf estimation iterations	UniformInt $[1, 20]$
ExtraTrees	Max features	$\{$sqrt, $0.5, 0.75, 1.0\}$
	Min samples split	LogUniformInt $[2, 32]$
	Min impurity decrease	Choice $\{0, 10^{-5}, 3 \cdot 10^{-5}, 10^{-4}, 3 \cdot 10^{-4}, 10^{-3}\}$

Computational resources The experimental evaluations were run on a CPU of 32 cores with an additional GPU for models that require GPU computations. Jobs were dynamically scheduled across nodes, so no pipeline ran exclusively on a given accelerator; runtime differences between model families span orders of magnitude (log scale in Figure 4a), so hardware heterogeneity is negligible relative to algorithmic differences.

GPUs: NVIDIA V100 (32GB VRAM), A100 (40GB VRAM), A40 (48GB VRAM)

CPUs: AMD EPYC 7742 64-Core Processor, AMD EPYC 7702 64-Core Processor (512GB RAM), Intel(R) Xeon(R) CPU E5-2660 v2, Intel(R) Xeon(R) Gold 6226R CPU (256GB RAM)

The total CPU and GPU hours for the entire STRABLE experiment is of 842 days.

Table D.2: Summary of Encoders and Architectures used in the STRABLE benchmark. Embedding dimensions, parameters and context length are taken from the English Massive Text Embedding Benchmark Leaderboard [11]

Category	**Model Name**	**Dim (d)**	**Params**	**Context**
Statistical Baselines	StringEncoder (Tf-Idf + SVD)	-	N/A	N/A
	TargetEncoder	-	N/A	N/A
	CatBoostEncoder	-	N/A	N/A
	FastText	300	N/A	N/A
Embedders (Contrastive)	E5 (Small/Base/Large)	384–1024	33M–335M	512
	BGE (Small/Base/Large)	384–1024	0.1B–33.5B	512
	UAE-Large	1024	335M	512
	All-MiniLM (L6/L12)	384	22M–33.4M	256
	All-MPNet-Base-v2	768	110M	384
	KALM (Embed)	896	0.5B	512
	Tarte	768	25M	N/A
Encoder-only (MLM)	RoBERTa (Base/Large)	768–1024	125M–355M	512
	DeBERTa-v3 (XS/S/B/L)	384–1024	22M–304M	512
	ModernBERT (Base/Large)	768–1024	149M–395M	8192
Encoder-Decoder	Sentence-T5 (Base/L/XL/XXL)	768	0.5B–5B	512
Decoder-only (Causal)	LLaMA-3.1 / 3.2	2048–4096	1B–8B	128k
	LLaMA-Nemotron-Embed-1B-v2	2048	1B	128k
	Qwen-3 (0.6B/4B/8B)	1024–4096	0.6B–8B	32k
	OPT (0.1B to 6.7B)	768–4096	125M–6.7B	2048
	Gemma-0.3B	768	300M	128k
	F2LLM (0.6B/1.7B/4B)	1024–2560	0.6B–4B	1024
	Jasper-0.6B	2048	0.6B	2048
End-to-End Architectures	ConTextTab	768	172M	256
	TabSTAR	384	47.2M	512

E Extended results

Take-aways on benchmarking tabular learning with strings:

- Strings carry signal that complements numerical features: every learner benefits from including string columns
- Real-world string columns are mostly short and repetitive (median 17 characters), not free text
- Modular pipelines outperform end-to-end string-tabular architectures
- Larger LLMs help only when paired with weak learners, or with appropriate post-processing of their embeddings, but they are not Pareto optimal (encoders are responsible for runtime)
- Decoder-only LLM embeddings need standard scaling or Matryoshka-style direct slicing rather than default PCA
- 30 principal components suffice; higher dimensions hurt performance and inflate runtime
- String length (avg. words per cell) is the dominant driver of ranking shifts; large LLMs only enter the top-10 on free-text-dominant tables
- STRABLE's 108 datasets yield rankings close to the oracle ($\tau \approx 0.93$), stable across application fields and data-preparation choices

Table E.2: Per-dimension variance concentration (Gini coefficient [10]) of embeddings, computed per dataset and aggregated across 108 datasets. **Higher Gini indicates that variance is more concentrated in a small subset of dimensions.**

Model	Median Gini	Mean Gini
MiniLM-L6-v2 (encoder)	0.152	0.164
E5-base-v2 (encoder)	0.113	0.126
BGE-large (encoder)	0.123	0.130
Qwen3-8B (decoder, Matryoshka)	**0.236**	**0.242**
OPT-6.7B (decoder)	**0.362**	**0.357**
LLaMA-3.1-8B (decoder)	**0.422**	**0.407**

Table E.3: **Pairwise comparisons of Gini coefficients across 108 STRABLE datasets.** Paired Wilcoxon signed-rank test [66] on Gini values per dataset, with p-values Holm-corrected for 15 pairwise comparisons [26]. H_0 : the paired differences in Gini Coefficients are different from zero. All decoder-vs-encoder comparisons are significant at $p < 10^{-17}$. The only marginal pair is E5-base-v2 vs BGE-large, two encoders with very similar Gini distributions.

Model A	Model B	Holm-corrected p
LLaMA-3.1-8B	MiniLM-L6-v2	2.8×10^{-18}
LLaMA-3.1-8B	E5-base-v2	2.8×10^{-18}
LLaMA-3.1-8B	BGE-large	2.8×10^{-18}
Qwen3-8B	MiniLM-L6-v2	2.8×10^{-18}
Qwen3-8B	E5-base-v2	2.8×10^{-18}
Qwen3-8B	BGE-large	2.8×10^{-18}
OPT-6.7B	MiniLM-L6-v2	2.8×10^{-18}
OPT-6.7B	E5-base-v2	2.8×10^{-18}
OPT-6.7B	BGE-large	2.8×10^{-18}

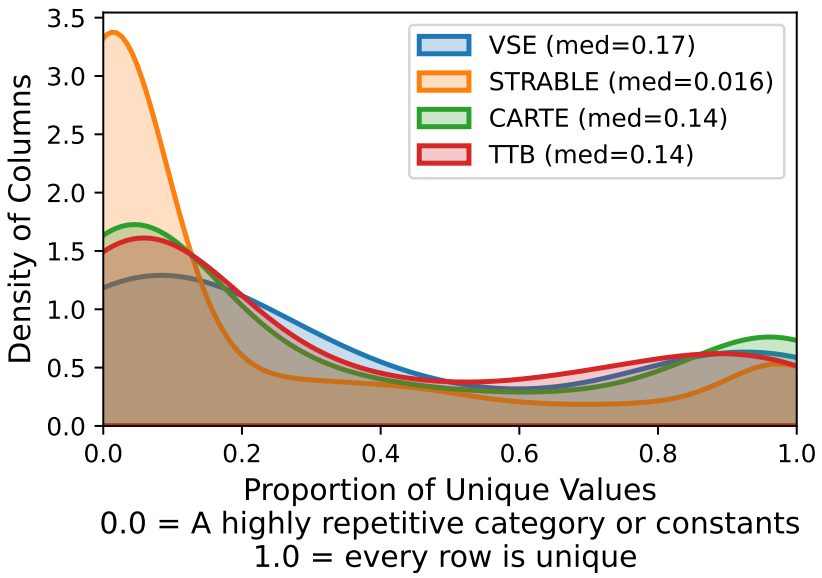


Figure E.1: VSE stands for the datasets used in 23, CARTE represents the datasets used in 32 and TTB represents the datasets used in TextTabBench [43]. STRABLE refers to the datasets introduced in this work. **STRABLE exhibits the lowest median proportion of unique text entries (0.016)**, compared to TTB (0.137), CARTE (0.136) and VSE (0.17).

- **Avg Tokens / Cell**: average number of whitespace-separated tokens per cell within a sample of 1000 rows.
- **Avg Char / Cell**: average number of characters per cell for a sample of 1000 rows.
- **Avg Unique Alphabetic Words / Cell**: average number of unique, case-insensitive alphabetic sequences (length ≥ 2) per cell, computed over a random sample of 1000 rows. This excludes numbers, punctuation, and single-letter characters.
- **Avg Unique N-grams / Cell**: average number of unique character n-grams (length 2–4) per cell within a sample of 1000 rows.
- **Proportion of Unique Values**: the ratio of unique values to total values in a column, indicating how repetitive vs. unique the entries are. **0.0** means all values are the same (e.g., a column of constants), while **1.0** means every value is unique (e.g., unique IDs or rich text).
- **Text Col Ratio**: proportion of columns that are text.

STRABLE shows the lowest metrics. In this regime, simple frequency-based methods such as Tf-Idf are well-suited and sufficient, as the discriminative signal is concentrated in a small set of recurring tokens rather than in semantic context — which explains the strong performance of Tf-Idf on the Pareto plot.

Table E.1: Median structural characteristics of text columns across the four benchmarks.

	Avg tokens / Cell	Avg char / Cell	Avg unique words / cell	Avg unique n-grams / cell	Prop. unique text entries	Text col ratio
CARTE	1.78	13.54	2.00	34.28	0.14	0.29
STRABLE	**1.31**	**11.07**	**1.35**	**26.81**	**0.02**	**0.59**
TTB	2.89	21.26	2.92	55.57	0.14	0.10
VSE	1.56	13.89	1.98	35.33	0.17	0.33

Table E.4: Comparison of score variations and runtime. The median score deltas (Δ) indicate a performance drop when shifting from 30 to 60 or 120 components, accompanied by a consistent increase in processing time.

Comparison	**Score Δ (Median)**	**Runtime**
60 vs 30	-0.0105	1.26x slower
120 vs 30	-0.0014	1.37x slower

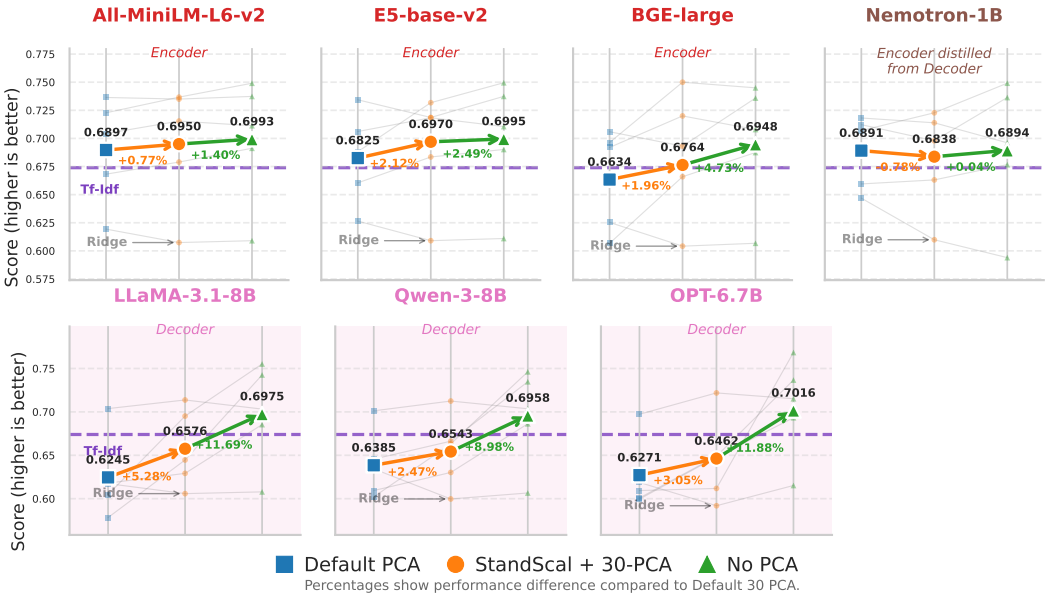
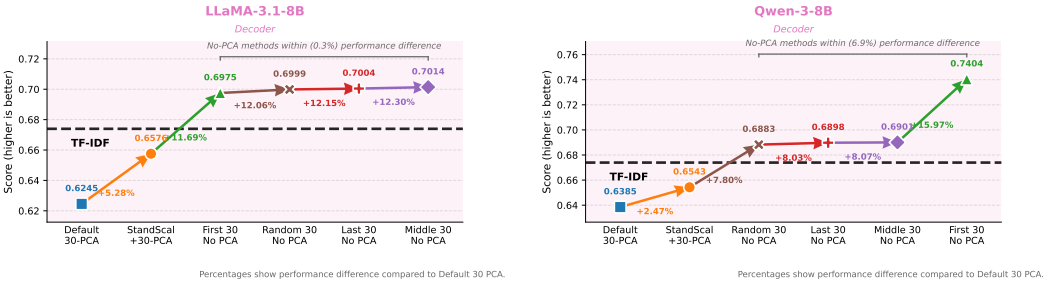


Figure E.2: **Per-learner breakdown of post-processing strategies across LM encoders.** Same setup as Figure 2, but with each individual learner shown as a faint background line (Ridge, XGBoost, ExtraTrees, TabPFN-2.5, TabICLv2) in addition to the across-learner mean (foreground). Top row: encoder-only models and the distilled Nemotron-1B. Bottom row: decoder-only models. The post-processing trend is consistent across learners — removing PCA helps decoders and is roughly neutral for encoders — with Ridge (annotated by arrows) consistently the weakest learner across all encoders and post-processing variants. The spread of background lines indicates that the choice of learner has a smaller effect on score than the choice of post-processing for decoder-only models, while the opposite holds for encoder-only models.



(a) Post-processing methods for LlaMA-3.1-8B

(b) Post-processing methods for Qwen-3-8B

Figure E.3: **Effect of embedding post-processing on LLaMA-3.1-8B and Qwen3-Embedding-8B.** Scores are averages of R^2 (regression) and AUC (classification) across all STRABLE datasets, over five learners (Ridge, XGBoost, ExtraTrees, TabPFN-2.5, TabICL). The dashed line marks the TF-IDF baseline. Percentages are relative improvements over Default 30-PCA. **Left (LLaMA-3.1-8B, decoder-only):** all four No-PCA slicing strategies (First/Last/Middle/Random 30) cluster within 0.3% of each other and consistently outperform both PCA variants by ∼12%, confirming that the choice of *which* 30 dimensions to retain is immaterial for this model. **Right (Qwen3-Embedding-8B, MRL-trained):** the four No-PCA strategies spread over 6.9%, with First-30 yielding a +15.97% gain—reflecting the model's MRL training, which explicitly front-loads semantic information into the leading dimensions. Qwen3-Embedding-8B's Matryoshka training supports a minimum of 32 dimensions; retaining 32 instead of 30 shifts the median score of Qwen-3-8B + TabPFN-2.5 from 0.786 to 0.788, with 104 of 108 tables within 0.01. *Note:* these results use unoptimized LLM inference.

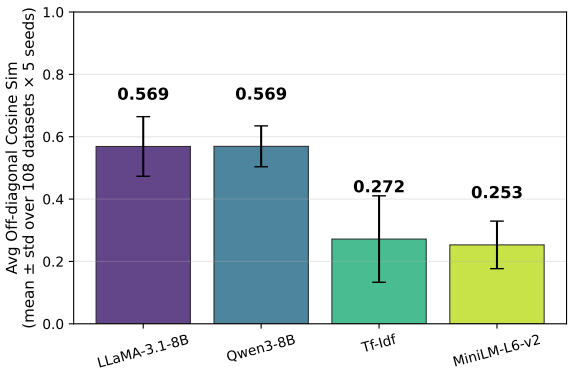
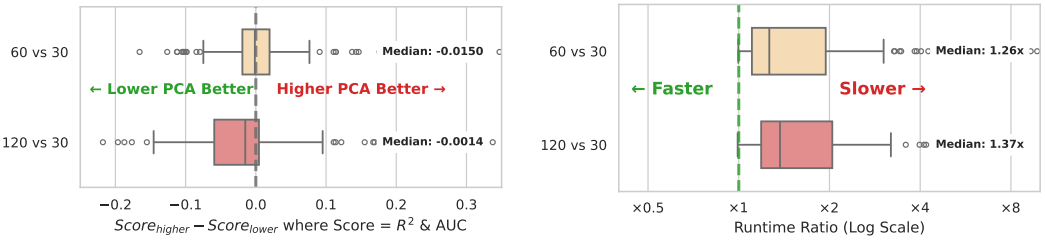


Figure E.4: **Decoder-only models LLaMA-3.1-8B and Qwen-3-8B exhibit notably higher average similarity (≈0.57) than encoder models such as MiniLM-L6-v2 (≈0.25), indicating that unrelated strings tend to receive similar embeddings.** Average off-diagonal cosine similarity of raw string representations across the 108 STRABLE datasets (5 seeds each). Tf-Idf is shown as a baseline.



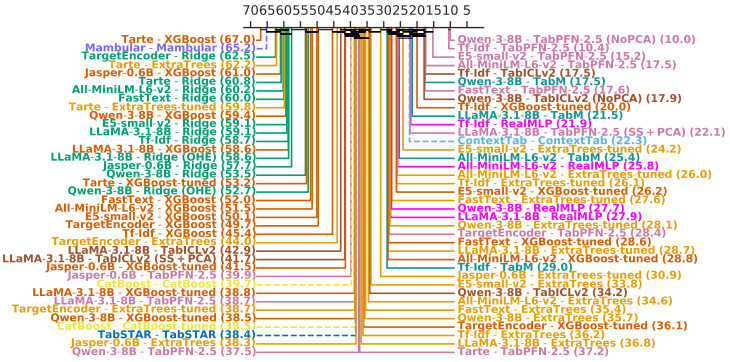
(a) Score deltas

(b) Runtime ratios (log scale)

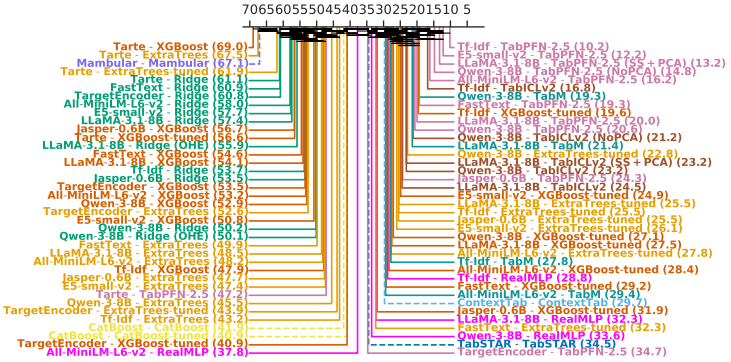
Figure E.5: **Performance difference between LLaMA-3.1-8B + TabPFN-2.5 with PCA at 30, 60, and 120 dimensions. Left:** distribution of per-dataset score deltas ($\text{Score}_{\text{higher}} - \text{Score}_{\text{lower}}$, where Score = R^2 & AUC). A negative median indicates that the higher-dimensional PCA hurts performance. **Right:** runtime ratio on a log scale; values > 1 indicate slower execution. Both comparisons use the 30-component setting as baseline. **30 components sit at a shallow optimum:** a 98%-explained-variance criterion retains a median of ∼200 components and drops the median score from 0.610 to 0.243, while 15 components lower it from 0.698 to 0.669 (15-PCA matches or exceeds 30-PCA on only 40/108 tables).

Table E.5: **Preprocessing preserves the Tf-Idf-vs-LLM gap.** For each dataset × learner × LLM encoder, gap = score(Tf-Idf) − score(LLM) is computed under the default and the alternative policy; $\Delta\text{gap} = \text{gap}_{\text{alt}} - \text{gap}_{\text{default}}$. We report the median and maximum of $|\Delta\text{gap}|$ across datasets (R^2 / AUROC).

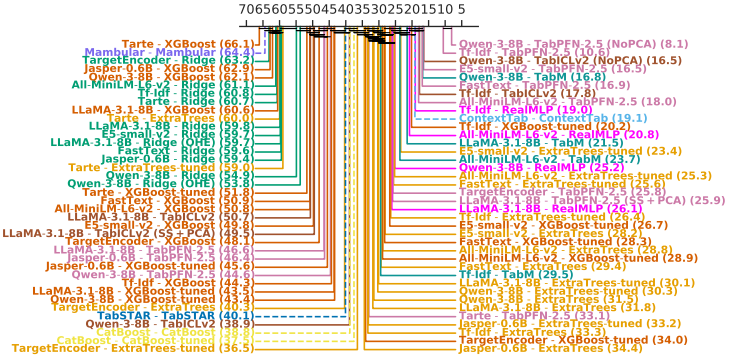
Alternative policy	median $\lvert\Delta\text{gap}\rvert$	max $\lvert\Delta\text{gap}\rvert$
Raw targets (no skew transform)	0.005	0.030
Manual feature engineering	0.001	0.010
Missing-value imputation	0.001	0.006
Subsampling (75K vs. full)	0.001	0.028



(a) CD Diagram for all tasks



(b) CD Diagram for Classification tasks



(c) CD Diagram for Regression tasks

Figure E.6: **Critical Difference (CD) Diagrams across different task types.** Comparison of all pipelines using the Friedman test. The diagrams show mean ranks and groups of models that are not significantly different (connected by horizontal bars).

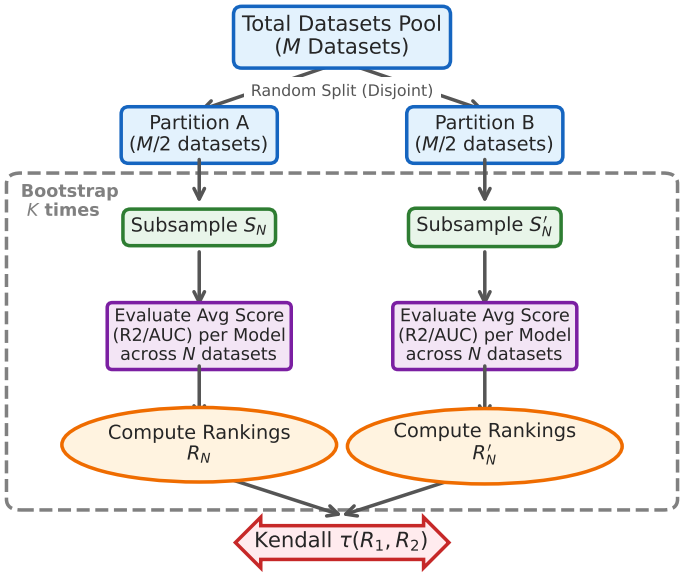


Figure E.7: Sampling Diagram to produce the boostrap estimator of the Kendall-τ correlation between independent benchmarks of size N

Table E.6: **TabuLa-8B underperforms all other pipelines.** `TabuLa-8B` [16] serializes each row and predicts its label autoregressively; at ≈2.4 s per test row, we evaluate it only on five small classification tables (`orphan-designations`, `paediatric-investigation-plan`, `tobacco-problem`, `drug-shortages`, `food-enforcement`). Average AUROC is reported. `TabuLa-8B` is pretrained on the T4 corpus discussed in section D.1, and its reported generalization has been largely attributed to train–test overlap [20]. Prompted or instruction-tuned generative LLMs are not evaluated, as they add configuration choices (serialization, prompting, shot selection) and are costly at benchmark scale.

Pipeline	AUROC
Qwen-3-8B + TabPFN-2.5	0.871
Tf-Idf + TabPFN-2.5	0.865
LLaMA-3.1-8B + TabPFN-2.5	0.861
ConTextTab (E2E)	0.860
TabSTAR (E2E)	0.832
CatBoost (E2E)	0.813
Mambular (E2E)	0.739
TabuLa-8B	0.614

Table E.7: Score differences between one-hot and passthrough encoding of categoricals are negligible for XGBoost and TabPFN-2.5. $\Delta = \text{score}_{\text{passthrough}} - \text{score}_{\text{OHE}}$ is computed per aligned (dataset, encoder) pair. $\overline{|\Delta|}$: mean absolute score difference across pairs. $\max|\Delta|$: largest absolute score difference observed. **% within 0.01**: fraction of pairs for which $|\Delta| < 0.01$.

Learner	n	$\overline{\lvert\Delta\rvert}$	$\max\lvert\Delta\rvert$	% within 0.01
XGBoost	0.007	0.083	80%
TabPFN-2.5	0.003	0.060	95%

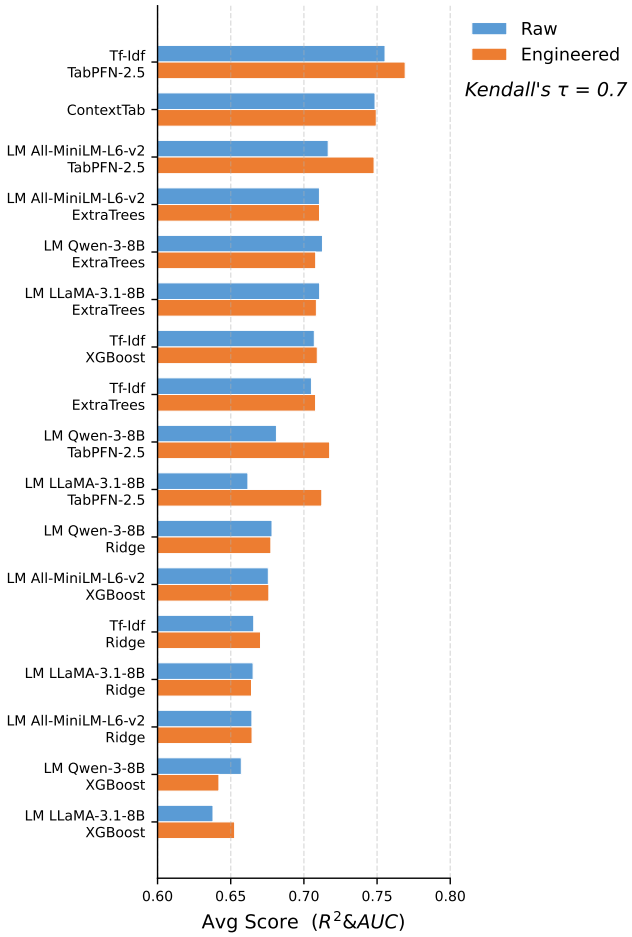


Figure E.8: **Raw vs. engineered features.** Average score for three representative pipelines with and without manual feature engineering applied to 44/108 tables (date parsing, ordinal encoding, range extraction, coordinate extraction, drug-strength parsing, fiscal year/quarter parsing). The Kendall-τ between the two rankings is reported.

Table E.8: Average score per learner by feature type.

Learner	Num-only	Num+Str	Str-only
CatBoost	0.475	0.695	0.624
CatBoost-tuned	0.462	0.694	0.617
ConTextTab	0.464	0.729	0.677
ExtraTrees	0.430	0.691	0.641
ExtraTrees-tuned	0.484	0.705	0.650
Ridge	0.328	0.623	0.580
TabPFN-2.5	0.485	0.686	0.585
TabSTAR	0.430	0.701	0.653
XGBoost	0.464	0.646	0.579
XGBoost-tuned	0.485	0.696	0.653
Average	**0.451**	**0.687**	**0.626**

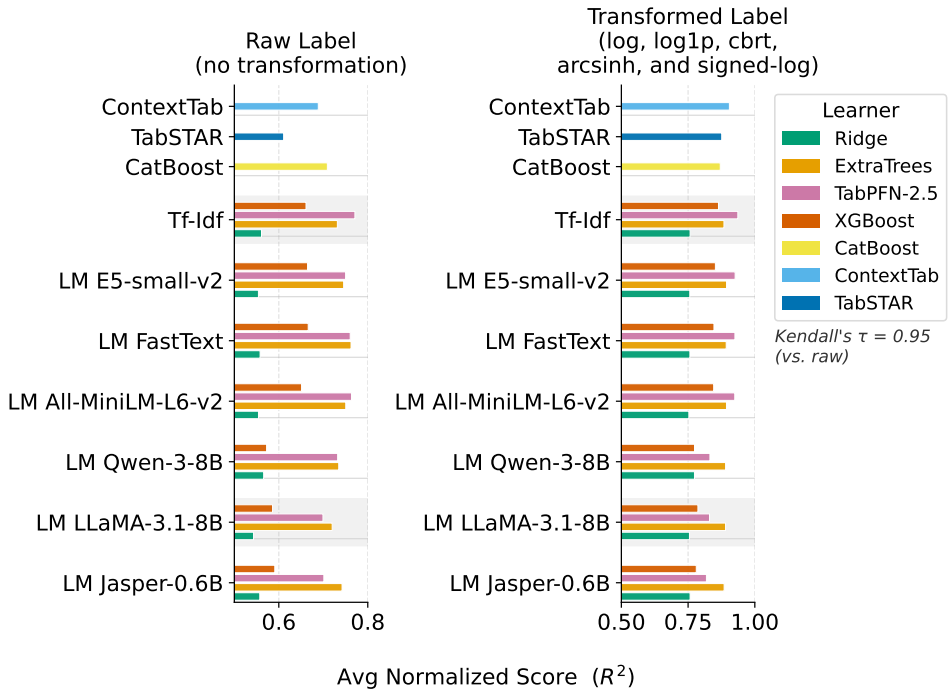


Figure E.9: **Impact of Label Transformation on Model Rankings**: We evaluated 31 pipelines (28 modular and 3 end-to-end) across 61 regression tasks. The bar charts compare average normalized R^2 scores using raw targets versus transformed targets. While models perform worse on raw targets due to unmitigated skewness and outliers, the relative performance and ranking of the encoders remain consistent across both settings.

Table E.9: Domain-level string meta-features and ranking stability (τ). τ: Kendall's tau measuring encoder ranking stability (higher = more stable). **n**: number of datasets in the category. **Words/Cell**: average number of words per cell. **Uniqueness**: ratio of unique values to total entries. **Vocab Div.**: ratio of unique words to total words across the category's string columns (lower values indicate more repetitive vocabulary).

Category	τ	**n**	**Words/Cell**	**Uniqueness**	**Vocab Div.**
Food	0.250	6	4.708	0.300	0.113
Education	0.475	10	5.824	0.354	0.145
Commerce	0.616	5	5.617	0.341	0.157
Social	0.693	4	2.035	0.073	0.035
Energy	0.702	9	2.894	0.219	0.085
Infrastructure	0.767	18	2.935	0.241	0.105
Economy	0.792	26	3.380	0.169	0.066
Health	0.834	30	4.913	0.256	0.083

Table E.10: Pareto Optimality Table

Encoder	**Learner**	**Score**	**Runtime (s/1k)**
TargetEncoder	Ridge	0.6593	0.0924
TargetEncoder	ExtraTrees	0.7290	0.1695
Tf-Idf	ExtraTrees	0.7407	0.9679
LM E5-small-v2	ExtraTrees	0.7422	2.5152
Tf-Idf	TabICLv2	0.7799	2.9988
Tf-Idf	TabPFN-2.5	0.7891	6.6700

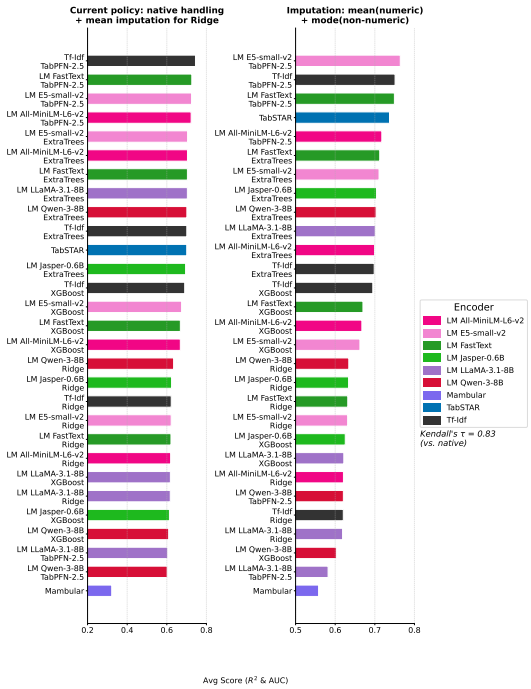


Figure E.10: **Impact of Missing Values imputation on Model Rankings.** Comparison of pipeline rankings under two missing-value strategies: native handling per learner with mean imputation for Ridge only (left), and mean/mode imputation applied to all pipelines before encoding (right). Bars show average score (R^2 & AUC) across 108 datasets, color-coded by encoder. Kendall's $\tau = 0.83$ between the two rankings, indicating that pipeline ordering is largely preserved regardless of imputation strategy.

Table E.11: Summary statistics for the number of columns per dataset after applying 30-PCA. **TabICLv2 was trained on a maximum of 100 features, while TabPFN was trained on up to 2000**. This discrepancy in the training feature budget may explain why TabICLv2, despite being highly competitive, performs slightly worse than TabPFN in this high-dimensional regime.

Statistic	Value
Count	108.00
Mean	416.17
Std	254.61
Min	95.00
25%	243.50
50%	344.50
75%	520.5
Max	1270.00

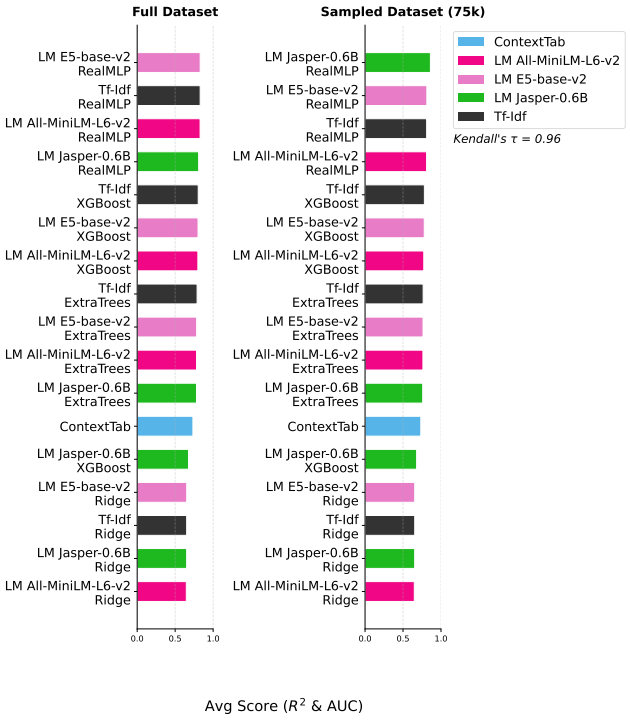


Figure E.11: **Pipeline rankings on full versus subsampled datasets** ($n = 75{,}000$)**.** The cap is applied to all pipelines (subsection C.3). To verify that it does not affect our findings, we re-run a subset of pipelines at full size on eight of the twelve datasets that exceed the cap ($n \in \{77{,}213; 80{,}358; 109{,}766; 117{,}984; 150{,}346; 170{,}730; 183{,}960; 270{,}009\}$): 17 pipelines ({Tf-Idf, All-MiniLM-L6-v2, E5-base-v2, Jasper-0.6B} $\times$ {Ridge, XGBoost, ExtraTrees, RealMLP}) and one end-to-end pipeline (ConTextTab). The rankings agree with Kendall-$\tau = 0.96$. Across all evaluated learners, the relative ordering of pipelines is preserved, and Tf-Idf consistently outperforms All-MiniLM-L6-v2 — suggesting that Tf-Idf's advantage in our benchmark reflects the nature of the strings in STRABLE rather than an artifact of subsampling.

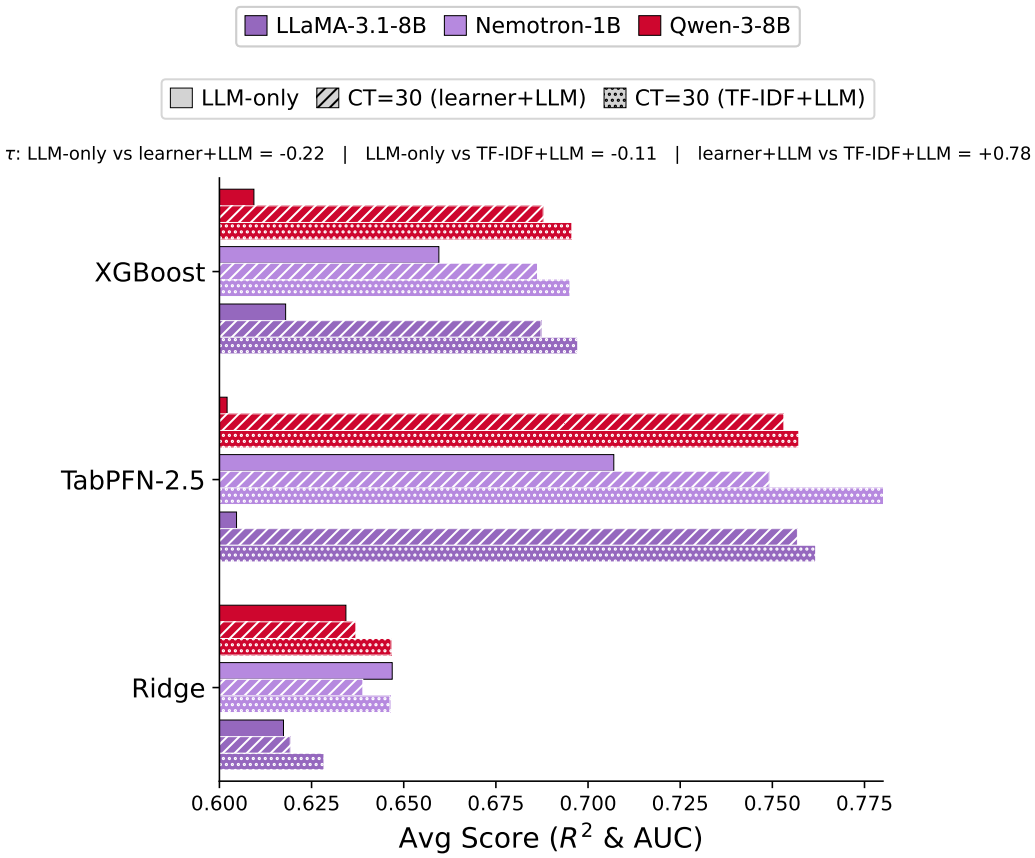


Figure E.12: **Effect of low-cardinality routing strategy on LLM encoders.** Average score (R^2 & AUC) for three encoders (LLaMA-3.1-8B, Qwen3-8B, Nemotron-1B) across three learners under three CT=30 routing strategies: **LLM-only** — all string features encoded by the LLM regardless of cardinality;**CT=30 (learner+LLM)** — low-cardinality features handled natively by the learner (XGBoost, TabPFN-2.5) or via OHE (Ridge), high-cardinality by the LLM; **CT=30 (TF-IDF+LLM)** — low-cardinality features ($\leq$30) handled by TF-IDF, high-cardinality by the LLM. Kendall's τ between the *TF-IDF+LLM* and *learner+LLM* rankings is $+0.78$, indicating that TF-IDF and native learner handling are largely interchangeable for low-cardinality features. By contrast, $\tau = -0.22$ between *learner+LLM* and *LLM-only*, and $\tau = -0.11$ between *TF-IDF+LLM* and *LLM-only*, showing that routing all features through the LLM disrupts rankings.

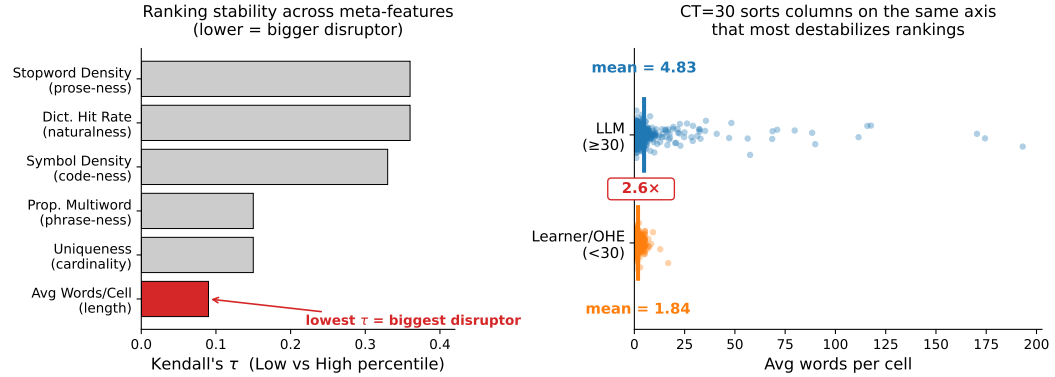


Figure E.13: **Longer text is redirected to LLMs encoding**. From figure E.12 we wanted to understand what kind of strings are encoded by the Language Models and what kinds are treated by the 30 Cardinality Threshold. As differentiating factor we use the average words per cell - being the string characteristic that most disrupts rankings. The picture shows that the average words per cell of features encoded by Language Models is 3 times more the length of features treated by OHE or by the native string handling of the learner.

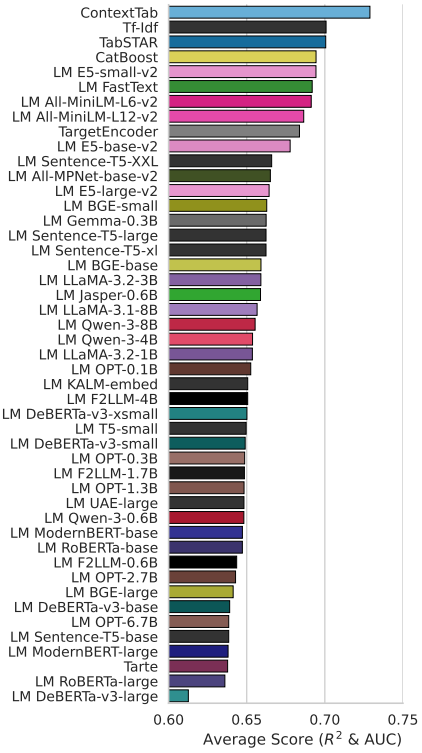


Figure E.14: Average performance per encoder. ConTextTab and TabSTAR performances - being end-to-end architectures - are based on their respective learners. Catboost encoder performance is based on the average between the default and tuned version of its respective learner. Seven encoders - All-MiniLM-L12-v2, E5-base-v2, E5-large-v2, LLaMA-3.2-1B, LLaMA-3.2-3B, Qwen-3-0.6B, Qwen-3-4B - were run fully on 4 learners - ExtraTrees, Ridge, XGBoost and XGBoost-tuned - and for 98% of the cases for TabPFN-2.5. Nine encoders, which are our representative ones - All-MiniLM-L6-v2, E5-small-v2, FastText, Jasper-0.6B, LLaMA-3.1-8B, Qwen-3-8B, TargetEncoder, Tarte, Tf-Idf - were fully run for ExtraTrees, ExtraTrees-tuned, Ridge, TabPFN-2.5, XGBoost, XGBoost-tuned. The remaining 28 encoders were run on ExtraTrees, Ridge and XGBoost.

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are listed in Table D.1; default-only configurations for TabPFN-2.5, TabSTAR, ContextTab, RealMLP, TabM, TabICLv2, and Mambular are explicitly stated. Preprocessing choices (PCA dimension, missing-value handling, target transformations, sub-sampling threshold) are detailed in Sections 3 and 4.

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Answer: [Yes]

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